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It’s Not What It Says, It’s How It Says It:
How Communication Design Shapes Human-AI Attachment

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Popular Science Summary

As AI chat systems become increasingly embedded in everyday life, the question of how they are designed to feel has become more important than ever. Some systems communicate in warm, empathising language and present themselves visually as friendly companions. Others are restrained, minimal, and task-focused. These differences are deliberate design choices, and this study examined what those choices actually produce in the people who use them.

Using a set of prototype AI chat interfaces that varied systematically in linguistic style and visual design, 129 participants evaluated how socially engaging, human-like, approachable, and trustworthy each felt. Two focus group sessions with ten participants then explored how people made sense of their responses and where they drew the line between acceptable and unacceptable AI design.

The findings revealed a consistent and striking tension. Warmer and more empathising design reliably produced stronger feelings of social connection, but simultaneously and without exception reduced perceived trustworthiness and credibility. Participants in the focus groups explained this through a sense that warmth in AI felt performed rather than genuine, a design strategy rather than authentic responsiveness. When an AI appeared to remember previous conversations without the user having initiated this, the majority found it intrusive rather than personal.

These findings suggest that designing AI systems to feel more like companions does not simply make people like them more. It makes people trust them less. These findings informed the development of Alongside, a speculative mobile application that explores what it means to design an AI system that redirects rather than accommodates relational needs, and what happens to user agency when it does. As AI becomes more deeply woven into social life, the most trustworthy system may not be the warmest one, but the most transparent one.

Keywords: AI Chatbots, Interface Design, Trust, Loneliness, Uncanny Valley, Social AI, Communication Design

Abstract

As AI chat systems become embedded in everyday life, the design choices that shape how these systems communicate have become increasingly consequential. Whether an AI uses empathising or neutral language, and whether it presents itself visually as a companion or a tool, are decisions with measurable effects on how users experience and relate to the system. As users increasingly turn to AI for social and emotional support, understanding whether the design choices that strengthen relational engagement are compatible with perceived trustworthiness has become both theoretically and practically urgent. Yet this relationship remains poorly understood.

To examine how linguistic style and interface framing shape relational perception in AI chat systems, this study employs an explanatory sequential mixed methods design combining research-through-design, a quantitative survey, and qualitative focus groups. Four scenarios were developed, each presenting two systematically varied prototype conditions, producing eight conditions evaluated by 129 survey participants. A subset of respondents then participated in two focus group sessions, where a separate set of four boundary-pushing design stimuli were used to probe the limits of acceptable AI design.

The findings reveal a consistent pattern in which empathising language and companion-oriented design reliably produced stronger relational responses while simultaneously reducing perceived trustworthiness across all eight conditions. Linguistic style emerged as the dominant variable, producing relational effects approximately twice as large as those of visual design. The study proposes a dual-process extension of the CASA paradigm and positions HAIA as the central theoretical framework for understanding relational attachment in AI contexts, with parasocial theory offering a complementary account of rapid relational activation. The study further delivers Alongside, a speculative design artefact that uses these findings to provoke reflection on consent, transparency, and the limits of responsible AI communication design.

Keywords: AI Chat Systems, Communication Design, Linguistic Style, Interface Framing, Parasocial Relationships, Human-AI Attachment, Trustworthiness, Relational Perception

Declaration of AI use

During the development of this thesis, AI-based tools were used to support specific and limited stages of the research and writing process. The tools used include Claude (Anthropic), ChatGPT (OpenAI), and Gemini (Google), all of which are also cited as design tools within the methodology for the generation of prototype response language.

These tools were used for four specific purposes. They supported grammar checking, spell checking, and stylistic consistency across sections of the thesis. They were used to brainstorm titles and structural ideas, summarise longer segments of text, and suggest alternative phrasings where needed. They contributed to the writing style of the AI response language used in the survey prototype conditions, where outputs from all three systems were generated, combined, and refined by the researchers to produce realistic AI interaction examples. Finally, they were used to support iterative refinement of thesis structure and section organisation during the writing process.

The role of these tools was supportive throughout, and their use raises considerations worth acknowledging. Where AI tools were used to suggest structural or stylistic changes, each output was critically reviewed and edited to ensure it reflected the authors' own reasoning and remained consistent with the theoretical framing and empirical analysis of the study. AI tools did not conduct analysis, interpret findings, or generate original argumentation. There is an inherent risk in using generative AI for writing support that certain phrasings or organisational choices may reflect the tendencies of the model rather than the authors' own voice, and this risk was mitigated through consistent critical review and revision of all AI-assisted outputs.

All theoretical framing, methodological decisions, empirical analysis, and written argumentation were produced by the authors. Their use was limited to appropriate tasks and conducted in accordance with the academic integrity standards of Malmö University. The authors understand the ethical implications of using AI in academic work and confirm that its use did not compromise the originality, authorship, or academic standards of this thesis.

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1. Introduction

In November 2022, ChatGPT reached one million users within five days of its public release, a milestone that took Facebook four and a half years to achieve (Kemp, 2025). By August 2025, the ChatGPT mobile application alone recorded 557 million monthly active users across Android and iOS, representing a 600% increase from two years prior and exceeding the platform's own web traffic figures (Kemp, 2025). Alongside ChatGPT, platforms including Google Gemini, Anthropic's Claude, and a growing range of specialised applications have become embedded in the daily lives of hundreds of millions of people worldwide. In the space of just a few years, AI chat systems, also known as large language models (LLM), have moved from being a novelty to being one of the most widely adopted communication technologies in human history. Yet the idea of machines that converse with humans is not new. The roots of conversational AI stretch back to the 1960s with the development of ELIZA, widely considered the world's first chatbot, whose creator Joseph Weizenbaum spent much of his later life warning about the psychological and social consequences of people forming emotional bonds with machines they mistakenly perceived as understanding them (Uenuma, 2026). That concern has gained renewed relevance as conversational AI systems have become embedded in the daily lives of hundreds of millions of users.

What makes this shift particularly significant is not simply the scale of adoption but how people are using these systems. Task completion, information retrieval, and writing assistance are well-documented use cases, but a less anticipated pattern has emerged alongside them. A nationally representative Pew Research Center survey of US teenagers found that 16% reported using LLM systems for casual conversation and 12% for emotional support or advice (Pew Research Center, 2025). A study published in JAMA Network Open found that one in eight US adolescents and young adults had sought mental health advice from a generative AI tool, with over 90% of those who did rating the experience as helpful (McBain et al., 2025). For a meaningful and growing share of users, AI chat systems do not represent only a productivity tool, they also function as a social outlet.

This development takes place against a broader social backdrop. In November 2023, the World Health Organization (WHO) declared loneliness a pressing global health concern, identifying it as one of the most underappreciated threats to human health and wellbeing globally. Users turning to AI chat systems for connection and emotional support is therefore not simply a product of technological novelty. It reflects something about the social conditions in which people are encountering these tools and the needs they bring to them. Recent research frames generative LLM systems as emerging affective social actors, marking a shift from purely functional tools toward systems that users increasingly interpret as companions or confidants (Zhong & Luo, 2025). As AI systems increasingly simulate relational qualities, understanding how emotional bonds with

them form has become both theoretically and socially important.

Unlike earlier automated systems, contemporary LLM-based chatbots can respond in real time, adapt to user input, and sustain extended exchanges that resemble interpersonal dialogue (Andersson, 2025). Through personalisation, continuity, and contextual awareness, these systems can create the impression of being listened to and socially accompanied in ways that earlier media technologies could not. This aligns with parasocial interaction theory, which argues that people can form perceived relationships based on communicative cues alone, even when interacting with entities that cannot engage in genuine reciprocal relationships (Horton & Wohl, 1956). In the context of conversational AI this dynamic becomes particularly complex, as the appearance of mutual exchange is considerably more convincing than in broadcast media, yet the underlying asymmetry remains the same (Maeda & Quan-Haase, 2024). The social experience the user has is real, while the reciprocity producing it is simulated.

The conditions of that simulation are, in significant part, a matter of design. Research consistently shows that specific communicative and visual choices shape how users perceive and relate to AI systems. Chaves and Gerosa (2021) identify empathy, self-disclosure, and personalised engagement as the characteristics most consistently linked to positive relational experience in human-AI interaction. Haugeland et al. (2022) demonstrate that conversational style shapes perceived anthropomorphism and hedonic quality. Norman (2013) and Ware (2008) establish that visual design communicates social intent before any exchange has taken place, meaning the relational character of an interface can be established before a single message is read. The CASA paradigm, which will be further discussed in section 2.2, demonstrates that even minimal social cues embedded in technology are sufficient to activate social scripts in users, producing relational responses that operate independently of whether users consciously acknowledge the system as a social entity. Design can therefore not be seen as neutral in the context of AI chat systems. It shapes what the interaction feels like and, by extension, what the system feels like as a social presence.

The AI chat landscape itself reflects this. General-purpose task-oriented systems such as ChatGPT, Gemini, and Claude sit at one end of a spectrum. Companion-oriented applications such as Replika and Character.AI, explicitly designed around emotional availability, persistent memory, and relational engagement, sit at the other. Between these poles lies a growing range of systems that combine functional capability with varying degrees of social and relational design. The communicative and visual choices that distinguish these systems from one another, including the tone and register of their language, the warmth or restraint of their visual identity, and whether they frame themselves as tools or companions, are design decisions with consequences for how users experience and relate to the systems they interact with, and for whether those systems are

perceived as credible and reliable (Skjuve et al., 2021; Brandtzaeg et al., 2022).

Which of these choices matter most, and what happens when the pursuit of relational warmth conflicts with the expectation of trustworthiness, has not been systematically established and forms the central concern of this study. Beyond its empirical contributions, the study produces a design artefact in the form of a speculative mobile application, developed in response to the study's findings and described in full in section 6.

1.1 Research Gap

Despite these advances, current research remains fragmented in systematically connecting specific communicative and interface design variations to users' relational perception of AI chat systems. Much of the existing literature either models predictors of usage intention, conceptualises attachment broadly, or examines individual cue types in isolation. There is limited research that adopts a research-through-design (RtD) approach to deliberately vary linguistic style and interface framing, meaning the visual and structural presentation of a system and how it positions itself in relation to the user, and empirically test how these variations shape relational experience as a media design outcome.

Equally absent is systematic investigation of whether the design choices that strengthen relational perception carry costs for perceived credibility and trustworthiness, a tension with direct implications for how AI communication systems are designed and evaluated. Klein (2025) points in this direction, noting that reliable functioning is a precondition for realising the relational benefits of human-like design, and Brunswicker et al. (2025) similarly identify a tension between empathising language and perceived credibility. Yet both confine their scope to individual cue types, and neither addresses what happens when linguistic style and interface framing work in opposing directions, or how relational warmth and reliability trade off against one another across systematically varied design conditions. That question has not been addressed.

1.2 Problem Formulation

A design-oriented understanding of how linguistic style and interface framing influence users' relational perception of LLM systems is currently missing from the literature. Existing theories, such as parasocial attachment, suggest that these design choices can shape whether AI is perceived more as a tool or a companion. However, it remains unclear which of the two design elements has the strongest influence on users' perceptions, how users respond when these elements are combined, and how they affect trustworthiness and credibility.

1.3 Relevance

Clarifying these relationships is relevant both theoretically and practically. Theoretically, the study contributes to extending parasocial and human-AI attachment frameworks into interactive AI contexts, where the conversational and responsive nature of LLM-based systems produces relational dynamics that differ meaningfully from the broadcast media environments in which these theories were originally developed (Horton & Wohl, 1956; Cong et al., 2026). In particular, it addresses a gap in understanding how the dual processes of relational activation and credibility evaluation operate simultaneously in human-AI interaction, a dimension that existing frameworks do not fully account for.

Practically, the findings are relevant for designers and developers making decisions about linguistic tone, interface framing, and relational cues in AI systems. As companion-oriented and task-oriented AI applications continue to proliferate, understanding which design choices strengthen relational experience, which undermine credibility, and how these effects interact has direct implications for responsible and user-centred AI design. This study is also relevant for policymakers and researchers concerned with the broader social consequences of AI systems that are increasingly embedded in users' everyday emotional and social lives, particularly given the WHO's identification of loneliness as a global health concern and the growing evidence that users turn to AI for social and emotional support. If the design choices that foster relational engagement are not well understood, systems risk deepening dependency on AI at precisely the moment when human connection is most needed.

1.4 Purpose & Objectives

This study aims to examine how linguistic style and interface framing in AI chat systems shape users' relational perception of those systems, including how those design choices relate to perceived credibility and trustworthiness. The purpose is to provide a deeper understanding of how specific communicative and visual design decisions influence whether users experience an AI system as a tool, a companion, or something in between, and what that means for how such systems are designed and evaluated. To fulfil this aim, the study employs a RtD approach (Zimmerman et al., 2007) combining a quantitative survey and qualitative focus group interviews, using a set of prototype AI chat interfaces that systematically vary in linguistic style and interface framing as the primary research instruments. The findings further inform the development of a design artefact in the form of a speculative mobile application, Alongside, which uses the study's conclusions to provoke reflection on consent, transparency, and the limits of responsible AI communication design.

1.5 Research Questions

The study is guided by one main research question and three sub-questions that address its constituent dimensions.

Main Research Question: How do variations in linguistic style and interface framing in AI chat systems influence users' relational perception of and engagement with those systems?

SQ1: How do relational linguistic cues affect users' perceived social engagement, trustworthiness, and human-likeness toward an LLM system?

SQ2: How does interface framing influence users' perception of an AI system as a social actor, and what effect does this have on perceived credibility and relational engagement?

SQ3: How do users interpret and make sense of their reactions to differently designed AI chat systems, and how do they relate these experiences to their broader social lives?

2. Literature review and Theoretical Framework

The following section presents the theoretical frameworks and prior research that underpin this study. Rather than examining these perspectives in isolation, the review develops a cumulative explanation of how users come to form relational perceptions of AI chat systems and how those perceptions may develop into longer-term attachment. Each framework contributes a different layer to this explanation, moving from the formation of one-sided relational bonds, to social responses to technology, to perceptions of human-likeness, and finally to the role of specific design features in shaping these processes.

The review begins with **parasocial interaction and parasocial relationships**, which establish that people can form meaningful relational bonds with mediated entities despite the absence of genuine reciprocity. It then turns to the **Computers are Social Actors (CASA) paradigm**, which explains why social cues embedded in technology automatically trigger interpersonal responses. The discussion of **anthropomorphism** builds on this by examining how users come to perceive technological systems as possessing human-like qualities. The fourth area focuses on the **design mechanisms** through which these responses are activated in AI contexts, with particular attention to linguistic style, interface framing, and relational cues.

The review culminates with **Human–AI Attachment (HAIA)**, the central theoretical framework of this study. Drawing on and extending the preceding perspectives,

HAIA explains how repeated interactions with AI systems may develop into enduring emotional bonds over time. Together, these frameworks provide the theoretical basis for understanding how communicative and interface design choices shape relational perception and form the foundation from which the study's theoretical propositions are derived.

2.1 Parasocial Interaction and Parasocial Relationships

The concept of parasocial interaction was introduced by Horton and Wohl (1956) to describe the one-sided sense of intimacy that audiences develop with media figures such as television personalities and radio hosts. Unlike genuine social relationships, parasocial interactions are non-reciprocal, as the media figure remains unaware of the individual audience member, yet the viewer experiences a perceived sense of connection and familiarity. Horton and Wohl (1956) observed that repeated exposure to the same media figure created the illusion of a growing relationship, as audiences came to feel they knew the on-screen persona personally. This was understood as a natural extension of human social behaviour into mediated environments, where people respond to social cues regardless of whether a genuine two-way relationship exists. Importantly, Horton and Wohl noted that this tendency was not inherently pathological or compensatory but reflected the deeply social nature of human beings and their capacity to extend relational meaning into mediated spaces, a point that later research has consistently supported.

At its core, parasocial theory rests on the observation that media figures address audiences in a direct and conversational manner, simulating the rhythms of ordinary interpersonal exchange. These communicative strategies, including direct address, inclusive language, and conversational pacing, trigger familiar social responses in viewers. The concept was foundational in media studies because it challenged the assumption that mass communication was passive and one-directional, demonstrating that audiences actively construct relational meaning from mediated content. Horton and Wohl (1956) identified the persona, meaning the mediated identity projected by the performer, as distinct from the performer's actual self, and argued that it was the persona's communicative style and relational consistency that audiences responded to. This observation carries significant implications for contemporary AI design, as it suggests that relational bonds form in response to communicative features rather than the perceived authenticity of a human being.

A later and important conceptual clarification was introduced by Dibble et al. (2016), who demonstrated that parasocial interaction and parasocial relationships, while related, refer to two distinct phenomena that had been routinely conflated in both theoretical and empirical work. Parasocial interaction refers specifically to the in-the-moment experience of feeling engaged with a media figure during an exposure situation. A parasocial relationship, by contrast, is a more enduring

socio-emotional bond that persists beyond individual exposure situations and builds through repeated contact over time. This distinction is relevant here, as the present study is concerned not with momentary in-session responses but with how design and linguistic choices shape the relational frame through which users perceive and engage with AI systems, which aligns with the domain of parasocial relationships as Dibble et al. (2016) define them. It is also worth noting that early characterisations of parasocial relationships as socially deficient have since been substantially revised, with later research reframing them as a normal and often beneficial feature of media engagement associated with reduced loneliness and meaningful emotional experience (Rubin et al., 1985).

While Horton and Wohl's (1956) framework was developed in the context of broadcast media, the concept has since been extended to digital environments where the relational dynamic becomes considerably more complex. In digital media the boundary between parasocial and social interaction begins to blur, as users can follow creators across platforms, engage with content in real time, and receive responses that feel personal even when they structurally are not. This shift challenges the classical model's defining feature of strict non-reciprocity, as digital environments produce conditions that simulate mutual exchange more convincingly than broadcast media ever could (Breves & van Berlo, 2025). Research on virtual influencers illustrates this development clearly. Breves and van Berlo (2025) conducted a four-week experience sampling study in which participants followed either a human or a virtual AI-generated influencer, finding that parasocial relationship intensity increased over time in both conditions and that participants progressed through comparable relational stages regardless of whether the influencer was human or virtual. These findings suggest that parasocial bond formation does not depend on perceiving the AI as human. Rather, it is the communicative and relational qualities of the interaction that appear to drive these connections. This implies that design features such as tone, responsiveness, and relational framing can independently contribute to the development of user attachment.

AI chat systems represent a further and more complex instantiation of this dynamic. Unlike virtual influencers, LLM-based chatbots generate adaptive responses in real time, adjust to user input, sustain conversational context, and personalise their language to individual users, creating a strong impression of genuine reciprocity. Maeda and Quan-Haase (2024), however, argue that this reciprocity is only perceived rather than actual. Because LLM systems generate language without intention, awareness, or social understanding, they cannot participate in genuinely reciprocal relationships. Instead, the sense of mutual exchange emerges from the system's ability to replicate conversational behaviours such as turn-taking, affirmation, and follow-up questions. The interaction is shaped in large part by the user's inputs and prompts, with the system producing responses that are statistically and contextually coherent rather than intentionally

caring, drawing equally on the patterns in the training data on which the model was built. It is on this basis that the parasocial framework remains applicable to human-AI interaction despite the surface appearance of reciprocity, as the bond, however strongly felt by the user, remains fundamentally one-sided. Recent research identifies emotional validation as a central mechanism in parasocial relationship formation with generative LLM systems, with anthropomorphism and adaptability contributing to perceived emotional responsiveness and social presence (Zhong & Luo, 2025).

2.2 Computers are Social Actors (CASA) Paradigm

The Computers are Social Actors (CASA) paradigm provides the most direct theoretical account of why users respond socially to technology even when they are consciously aware that the system they are interacting with is not human. The paradigm is rooted in the work of Reeves and Nass (1996), whose central argument is that the human brain lacks an evolved mechanism to automatically distinguish mediated representations from real-life counterparts. As a result, users respond to technologies in ways that are fundamentally social and instinctive, applying the same communicative scripts they would use in human-to-human interaction to their encounters with media and computing systems. The critical insight is that this happens automatically. Even minimal social cues are sufficient to elicit social responses, and technologies do not need to be fully human-like for users to engage with them as social entities. Nass and Moon (2000) confirm that these social responses are not the result of confusion about the nature of the technology but persist even among users who explicitly acknowledge they are interacting with a machine. Social responses to communicative cues, in other words, operate independently of and prior to conscious evaluation, a distinction that carries significant implications for how the findings should be interpreted.

The mechanisms underlying CASA effects have been debated across several explanations, each of which illuminates a different aspect of how design choices shape social perception. Nass (2004) argues that social attitudes and perceptions are evoked by specific social cues, identifying language, voice, emotional expression, involvement, and unpredictability as among the features that trigger these responses. Fiore et al. (2013) refine this by distinguishing between social cues, defined as physically and biologically determined features that function as channels of useful information, and social signals, which are the meaningful interpretations perceivers derive from those cues, including attributed emotions, personalities, and engagement. This distinction matters because it clarifies that what users respond to is not the raw feature itself but the meaning they derive from it, which means that design choices function as communicative acts whose relational significance is constructed in the moment of perception. Research building on this framework has consistently found that casual, informal, and warm language styles exert stronger influence on users' social perception of technologies than formal, cold, and objective language (Goble and Edwards, 2018; Hoffman et

al., 2020). This pattern positions linguistic register as one of the primary levers through which AI systems are experienced as more or less socially present.

A second explanatory account centres on mindlessness. Sundar and Nass (2000) propose that users unconsciously treat computers as if they were real people not through deliberate reasoning but through the automatic application of interpersonal rules learned through repeated social experience. Fischer (2011) develops this argument by suggesting that prolonged exposure to social cues in everyday human interaction creates cognitive scripts that are triggered automatically when those same cues appear in technological contexts, bypassing reflective evaluation. Nass and Moon (2000) confirm that this process operates regardless of users' stated attitudes toward technology, meaning that someone can simultaneously hold the conscious belief that AI is not a social entity and respond to it as if it were. As a result, the CASA paradigm highlights a potential distinction between users' stated attitudes and their actual responses to social design cues, a possibility with important implications for the study of human–AI interaction.

The most consequential extension of the CASA paradigm for this study concerns conversational agents. Edwards et al. (2019) found that participants responded to LLM systems in ways fully consistent with CASA predictions, attributing social presence, credibility, and relational qualities to them based on communicative cues alone. Language emerges as the central mechanism in AI interaction precisely because it is the primary channel through which users infer the presence of a social mind. Pizzi et al. (2023) extend this by suggesting that users evaluate LLM systems interactions through social dimensions such as warmth, competence, and credibility in ways that closely mirror how they evaluate human conversational partners, with real behavioural consequences including willingness to disclose personal information and continue the interaction. Xiao et al. (2019) identify interactivity as a further amplifier of CASA effects, arguing that the dynamic and reciprocal structure of LLM systems exchanges sustains social responses across the interaction rather than simply triggering them at the outset. Jung et al. (2014) and Liew and Tan (2018) indicate that these effects extend across interface formats, suggesting that social responses are activated by the communicative and visual character of an interface rather than by any specific technological platform. Much of this research, however, predates LLM-based systems, and whether these patterns hold in more sophisticated conversational AI remains an open question.

The way chatbots are framed within interfaces, whether as assistants, tools, or companions, adds a further layer to this picture. A meta-analysis of chatbot studies found that while social cues consistently increase user engagement, trust, and openness, their effectiveness is shaped by contextual and design factors including how the system positions itself in relation to the user (Klein, 2025). This points to interface framing as a design variable that operates on social perception not by

adding social cues directly but by establishing the interpretive context within which other cues are read. A warm language style encountered in a tool-framed interface may produce a different relational response than the same language encountered in a companion-framed one, because the frame primes the user's expectations before any content is processed.

What the CASA paradigm does not fully account for, however, is the possibility that the same design features which strengthen relational perception may simultaneously reduce perceived trustworthiness and credibility. The paradigm predicts that social cues activate social scripts, but it was developed in contexts where the primary concern was engagement and likeability rather than the reliability demands that users place on AI systems used for information, advice, or decision support. In such contexts, the activation of social scripts may be accompanied by a parallel evaluative process in which warmth is read as a signal of reduced objectivity. Understanding whether relational perception and credibility operate on different and sometimes opposing tracks requires theoretical resources that extend beyond CASA, which the subsequent sections begin to provide.

2.3 Anthropomorphism

Anthropomorphism describes the process through which individuals attribute human-like emotions, mental states, and intentions to nonhuman entities in order to interpret and make sense of their behaviour (Epley et al., 2007). Whereas the CASA paradigm emphasizes the automatic application of social responses to technology, anthropomorphism examines the attribution of human-like qualities to technological systems. Epley et al. (2007) propose that this process is driven by the accessibility of human knowledge schemas, the desire to predict the environment, and the need for social connection, making it more likely to occur when a system's behaviour feels uncertain or emotionally attuned. Critically, this process is not only a psychological tendency but a designable one. According to Schaumburg (2001), interfaces can be deliberately designed as social entities through choices such as naming, visual presentation, tone, and role assignment. This distinguishes anthropomorphism as a user-driven perception from anthropomorphic design as a strategy intended to evoke that perception. Xu et al. (2022) confirm that when systems display social cues such as casual language, emotional expression, or adaptive behaviour, users are more likely to interpret them as intentional and socially capable agents, with the degree of this attribution moderated by how reflectively users engage with the system rather than simply responding to it automatically.

What remains less clear in this body of research is whether the relational benefits of anthropomorphic design are always accompanied by positive evaluations in other dimensions. Existing studies largely emphasize outcomes such as engagement, empathy, social presence, and emotional connection (Li et al., 2025),

while giving less attention to how anthropomorphic cues may influence perceptions of credibility and trustworthiness. This gap may be particularly relevant in the context of contemporary LLM-based chat systems, which are capable of sustaining adaptive, highly conversational interactions and are increasingly used for both informational and relational purposes. As a result, further research is needed to understand how users balance social and evaluative responses to AI systems, and how these responses may contribute to the development of longer-term human–AI relationships. These questions are addressed by the Human-AI Attachment framework discussed in section 2.5.

2.4 Design Mechanisms in Human-AI interactions

The theoretical frameworks examined in the preceding sections establish why users respond socially to AI systems and how those responses develop into anthropomorphic perception. What they leave less specified is which particular design choices activate these processes. This section examines the two design variables that are central to the present study, linguistic style and interface framing, as the primary mechanisms through which social responses to LLM systems are shaped.

Language is the most direct and consistently evidenced mechanism through which social responses to AI systems are produced. Variations in tone, register, and conversational style influence how users perceive the personality, intent, and social character of a system before any conscious evaluation takes place. Research on conversational agents consistently shows that casual, warm, and human-like language styles are perceived as more socially engaging than formal, neutral, or task-oriented alternatives (Goble and Edwards, 2018; Hoffman et al., 2020; Nass and Moon, 2000). Chaves and Gerosa (2021) identify empathy, self-disclosure, and personalised engagement as the social characteristics most consistently linked to positive relational experience in human-AI interaction, establishing that users respond most strongly to language that acknowledges their emotional state, adapts to their conversational register, and addresses them as individuals rather than generic users. Haugeland et al. (2022) complement this by demonstrating experimentally that open-ended, topic-led conversational styles strengthen perceived anthropomorphism and hedonic quality compared to task-led approaches that prioritise efficient goal completion. Relational cues embedded in language, including personalisation, memory references, self-disclosure, and conversational reciprocity, further contribute to this effect by signalling attentiveness and social awareness in ways that mirror key features of human interaction (Liang et al., 2021; Skjuve et al., 2022).

Empathising language represents the most potent of these cues. Expressions of understanding, validation, and emotional support have been found to enhance perceived connection and emotional engagement with chatbots (Fitzpatrick and Darcy, 2017; Skjuve et al., 2022), and Xiang et al. (2025) demonstrate that relational

cues of this kind significantly increase emotional arousal, trust, and attachment in users. However, the relational benefits of empathising language do not come without cost. Brunswicker et al. (2025) demonstrate that empathising language in task-oriented conversational agents produces a systematic trade-off between perceived warmth and perceived credibility, with users rating empathising conditions as more socially engaging but less factually reliable than neutral alternatives. This trade-off having been demonstrated for linguistic style alone, without accounting for the role of interface framing or testing what happens when the two variables work in opposing directions, leaves the central design question open.

Interface framing influences how users interpret and engage with AI systems through the visual cues embedded in the interface. Norman (2013) argues that design features communicate a system's intended purpose and character before interaction begins, while Ware (2008) demonstrates that elements such as colour, typography, and layout shape users' impressions at an early stage of perception. In the context of AI chat systems, these visual cues can encourage users to view the system either as a functional tool or as a more socially oriented conversational partner.

A tool-oriented interface typically emphasizes efficiency, clarity, and task completion through restrained visual design and productivity-focused framing. In contrast, a socially oriented interface may incorporate warmer visual elements, conversational branding, and design features that encourage a sense of interaction and connection. According to Preece et al. (2019), these different design orientations can lead users to form different expectations about how they should engage with a system.

As such, interface framing may influence how users interpret LLM language. The same empathising response may be perceived differently depending on the visual context in which it appears. A socially oriented interface may reinforce the relational intent of the message, whereas a tool-oriented interface may encourage users to interpret the same message primarily as a functional interaction. In this way, interface framing shapes the context through which linguistic cues are understood. Klein's (2025) meta-analysis confirms this, finding that the effectiveness of social cues in LLM interaction is moderated by contextual and design factors rather than operating uniformly across conditions.

What remains less clear from the existing literature is what happens when linguistic style and interface framing work in opposing rather than complementary directions, and whether one variable exerts greater influence on relational perception than the other when they conflict. The literature provides limited understanding of the relative influence of linguistic style and interface framing on users' perceptions of AI chat systems. It also remains unclear how design choices

intended to foster relational warmth relate to perceptions of credibility and trustworthiness.

2.5 Human-AI Attachment

Human-AI Attachment (HAIA) is introduced by Cong et al. (2026) as a one-way, non-reciprocal emotional bond formed by individuals towards AI systems through direct interaction. While the concept draws on parasocial attachment theory, it is positioned as a distinct subtype occupying a conceptual space between traditional parasocial relationships, which involve no direct interaction with the media figure, and fully reciprocal interpersonal bonds. What distinguishes HAIA from parasocial relationships in particular is the nature of the exchange. Unlike a viewer's bond with a television presenter, HAIA develops through actual conversational interaction with the AI system, even though the AI remains without genuine awareness or emotional investment in the relationship.

This distinction positions HAIA as the framework most directly suited to understanding how the design of AI chat systems shapes relational outcomes, because it locates the mechanism of attachment formation in the interaction itself rather than in one-sided observation of a mediated figure.

Cong et al. (2026) propose a three-stage developmental model in which attachment progresses from functional expectation through emotional evaluation to the establishment of stable internal representations of the AI as a reliable and responsive presence. In the first stage, users approach an AI system with expectations shaped by prior experience with technology and by the anthropomorphic design of the interface. If those expectations are confirmed through early interaction, users progress to the second stage, in which the system's responsiveness to emotional and social needs generates positive affect and growing investment in the relationship. Over time, these experiences consolidate into the third stage, in which users develop stable internal working models of the AI that persist across interactions and influence future use. This model carries a direct practical implication for communication design: each stage is sensitive to different features of the system, meaning that design choices made in the earliest moments of interaction can shape the entire relational trajectory that follows.

While the HAIA framework was formalised only recently, earlier empirical research on AI companion systems anticipates many of its central claims. Skjuve et al. (2022) tracked users' relationships with the Replika chatbot over an extended period, finding that while early interactions are characterised by curiosity and exploratory self-disclosure, the relationship evolves toward greater emotional depth and stability as trust develops. Brandtzaeg et al. (2022) conducted in-depth interviews with Replika users who described the chatbot as a friend, finding that the relational

quality users attributed to the system was actively produced by specific features of its communicative design, including its ability to ask personal questions, remember prior conversations, and adapt its tone to the user's emotional state. Andersson (2025) situates these developments within a broader shift in how people engage with AI, arguing that users increasingly approach these systems not as tools but as relational presences, with expectations of engagement, memory, and emotional responsiveness that go beyond functional task completion.

The conditions under which HAIA develops are further specified by research on specific interaction features. Xiang et al. (2025) demonstrate that relational cues including empathy, self-disclosure, and conversational continuity significantly increase emotional arousal, trust, and attachment, but that these effects are moderated by the degree to which users attribute psychological capacities to the LLM systems. This suggests that relational outcomes are jointly determined by the designed features of the system and the perceptual frame users bring to the interaction, meaning that the same design choices may produce substantially different relational responses depending on users' prior orientation toward AI. Peng et al. (2025) support this by finding that anthropomorphic interface design moderates the relationship between parasocial interaction and continued AI use, with users who encounter more humanised interfaces being more likely to develop and sustain relational engagement over time. Smith et al. (2025) offer a useful counterpoint, arguing from a theoretical standpoint that while LLM interactions share surface features with close human relationships, they fall short of fulfilling the deeper functions of interdependence and genuine mutual responsiveness. This positions HAIA as a qualitatively distinct relational experience rather than a substitute for human connection.

For this study, the HAIA framework helps differentiate between immediate relational responses and the longer-term process through which attachment may emerge. Given that attachment develops over time, single-exposure interactions are unlikely to capture attachment itself. Instead, measures such as social engagement, human-likeness, approachability, and trustworthiness can be understood as indicators of the relational conditions that may contribute to attachment development through repeated interactions.

More broadly, HAIA provides the framework that integrates the preceding theories into a coherent account of human–AI relationship formation. Parasocial theory establishes that relational bonds can emerge in the absence of genuine reciprocity, CASA explains why users respond socially to technological systems, anthropomorphism addresses how those systems come to be perceived as human-like, and the design mechanisms literature identifies the communicative and interface features through which these processes are activated. HAIA extends these perspectives by explaining how such responses may develop into enduring emotional bonds over time. As such, it provides the central theoretical foundation

for examining how communicative and interface design choices shape the early stages of human–AI relationship formation and forms the basis for the theoretical propositions presented in the following section.

2.6 Theoretical Propositions

The propositions draw on the cumulative logic of the preceding framework. Parasocial theory, CASA, and anthropomorphism suggest that social and human-like cues influence relational perception, while HAIA proposes that these responses may form the basis of longer-term attachment processes. These are not formal hypotheses but design-oriented expectations derived from the existing literature, against which the study's findings are examined.

P1: AI chat systems using empathising and relational linguistic cues were anticipated to orient users toward stronger relational perception than systems using neutral or impersonal language.

P2: AI chat systems framed as companions were expected to be perceived more strongly as social actors and to produce stronger relational engagement than systems framed as functional tools.

P3: The combination of relational linguistic cues and companion-oriented interface framing was expected to produce the strongest relational engagement of all design combinations.

3. Method

To examine how linguistic style and interface framing in AI chat systems shape users' relational perception, this study employs an explanatory sequential mixed methods design combining a quantitative survey phase with a qualitative focus group phase (Creswell, 2015). The quantitative phase establishes measurable differences in how participants respond to systematically varied prototype AI chat conditions, while the qualitative phase explores how participants interpret and make sense of those responses from their own perspective. Neither method alone is sufficient, the survey produces the comparative patterns the study requires, while the focus groups provide the interpretive depth needed to understand what those patterns mean. The integration of the two phases follows a connected approach (O'Cathain et al., 2010), in which focus group participants are drawn directly from survey respondents and the group discussions are shaped by the patterns the survey identifies, ensuring the two strands of the study speak to each other rather than running in parallel.

The mixed-methods design was evaluated against quality criteria relevant to explanatory sequential research outlined by Creswell (2015). Integration quality was addressed by drawing focus group participants directly from the survey

sample and shaping discussions around the quantitative patterns the survey identified. Credibility in the qualitative phase was supported through independent coding, collaborative theme development, and the requirement that themes appear across both sessions. Construct validity in the quantitative phase was addressed through scenario-based stimulus design grounded in established theory, consistent perceptual measures across all four scenarios, and prior piloting of the survey instrument. Transferability is acknowledged as limited by the sample's demographic skew and cultural context, addressed in section 3.5.

This study applies Research through Design (RtD) (Zimmerman et al., 2007) as its core methodological approach, using the design and systematic variation of prototype AI chat interfaces as the primary means of operationalising the study's two independent variables, linguistic style and interface framing. These prototypes function as the research instruments through which both phases of data collection are conducted. In the first phase, participants rate the prototype conditions through an online survey across a set of perceptual dimensions, producing quantitative data about how the design variations affect perceived social engagement, approachability, human-likeness, and trustworthiness. In the second phase, a subset of survey respondents is invited to take part in two focus group sessions, which allow the patterns observed in the survey data to be examined in greater depth through a different set of prototype AI chat interfaces than in the survey and explored from participants' own perspectives.

The choice of this research design over alternatives was informed by the nature of the research questions and the study's practical constraints. Questions about how humans relate to AI systems resist clean solutions, and quantitative measurement alone cannot capture the full complexity of the relational, ethical, and social dynamics involved. Reframing these questions in ways that are accessible and open to reflection requires design work and qualitative inquiry alongside numerical comparison, which is why the involvement of both focus groups and prototype development was not simply a practical addition to the study but a methodological necessity. A purely quantitative experimental design conducted under controlled laboratory conditions could have offered greater internal validity but would have sacrificed the realism that the online survey format provides and made large-scale recruitment considerably more difficult within the study's timeframe. A qualitative-only approach using longitudinal diary studies or in-depth interviews would have produced richer data about individual relational experience over time but could not have generated the comparative perceptual patterns across conditions that the study required. A longitudinal design would be necessary if the study aimed to capture the full developmental trajectory that HAIA describes, but since this study is concerned with the conditions under which early-stage relational responses are activated rather than how attachment develops over time, a single-exposure design is appropriate to the scope.

3.1. Research through Design

Research through Design (RtD) was selected as the core methodological approach for this study because it makes it possible to isolate and manipulate specific design variables with a degree of precision that studying pre-existing systems would not allow (Zimmerman et al., 2007). Rather than observing how users respond to AI chat systems already available on the market, where linguistic style and interface framing are inseparable from brand identity, platform conventions, and prior user experience, this study required a controlled environment in which those two variables could be varied independently and in combination. RtD provided that environment by treating the design of artefacts not as an output of the research process but as a mode of inquiry within it, allowing the theoretical constructs identified in the literature review to be translated into concrete and testable prototype conditions.

Zimmerman et al. (2007) describe RtD as producing knowledge at three distinct levels: through the design process itself, through the knowledge embodied in the artefact, and through the knowledge generated by users' responses to it. All three are present here. Designing and iterating the prototype conditions produced insight into how linguistic and visual variables interact and how difficult it is to manipulate one without affecting the other. The prototypes themselves embody theoretical claims about what distinguishes empathising from neutral language and visually warm from visually restrained interface framing. The survey and focus group data then contribute empirical knowledge about how those distinctions shape relational perception in practice.

3.1.1 Design Variables & Process

The study examines two design variables: linguistic style and interface framing. Linguistic style refers to how the AI communicates, while interface framing refers to how the system is visually presented. These variables are manipulated across four scenarios, each comparing two versions of the same AI chat interaction, allowing for comparison between different design conditions. The conditions differ across these two design dimensions, varied independently or in combination depending on the scenario. Within linguistic style, empathising refers to language characterised by emotional acknowledgment, informal phrasing, and first-person social engagement, while neutral refers to language that is factual, objective, and task-oriented without acknowledging the emotional content of the user's message. Within interface framing, visually warm refers to interfaces characterised by soft colour gradients, organic visual motifs, and inviting greeting language, while visually restrained refers to interfaces characterised by minimal design, structured typography, and functionally worded interface elements. These descriptors are used in place of evaluative terms such as friendly, which can carry implicit connotations that may influence how the findings are interpreted.

Chaves and Gerosa (2021) identify empathy, self-disclosure, and personalised engagement as the social characteristics most consistently linked to positive relational experience in human-AI interaction. This establishes that users respond more relationally to systems that acknowledge their emotional state, adapt their register to the conversational context, and address them as individuals rather than as generic users. The empathising linguistic conditions in this study were constructed on this basis, with responses acknowledging the user's message emotionally before providing practical content, using informal and conversational phrasing, and in some scenarios referencing the user's prior interactions with the system. Haugeland et al. (2022) complement this by demonstrating experimentally that open-ended, topic-led conversational styles strengthen perceived anthropomorphism and hedonic quality relative to task-led approaches that prioritise efficient goal completion. The neutral conditions in this study are accordingly characterised by direct, structured responses oriented toward information delivery, while the empathising conditions adopt a more open and dialogic register that invites continued engagement.

The visual design decisions draw on two complementary bodies of design theory. Ware (2008) establishes that colour, form, and typography communicate meaning pre-attentively, before users consciously process written content, informing the overall visual character of each condition. Norman (2013) extends this through the concepts of affordances and signifiers, arguing that design features communicate intended use and social character before interaction occurs. The warm-toned gradients, organic motifs, rounded typefaces, and inviting input field text of the visually warm interfaces were designed to signal openness, social availability, and relational engagement. In contrast, the restrained colour palettes, structured typography, and functionally worded interface elements of the visually restrained interfaces were designed to signal efficiency and task-oriented purpose. These visual elements function as signifiers that shape users' relational interpretations before a single message is exchanged, making the differences between conditions substantive rather than purely aesthetic.

All prototype designs were developed as mobile app interfaces, reflecting the dominant context in which AI chat systems are now primarily used (Kemp, 2025). This ensures that participants encounter the prototypes in a format consistent with how they most commonly interact with LLM systems in everyday life. The mobile format also functioned as a productive design constraint, limiting the available visual space and encouraging fewer but more deliberate differences between conditions. The visual design choices draw on Western conventions of colour psychology and interface layout, as the study was distributed primarily to a Western European audience. This is a deliberate scope limitation rather than an oversight and is addressed further in the limitations section.

The prototype conditions described in this section served as the research instruments for the quantitative survey phase and established the design baseline from which a separate set of stimuli was developed for the focus group phase, where the aim shifted from measuring perceptual responses to controlled variables toward probing the limits of acceptable AI design. Those stimuli are described in section 3.3.

3.1.2 Prototype Development and Iterations

The development of the prototypes was iterative and involved several stages of design exploration, internal review, and refinement, with Figma used as the primary prototyping software throughout. The process began with a broad exploratory phase in which a wide range of interface concepts and scenario types were developed and examined, including designs featuring humanoid characters and abstract figures, experiments with typographically and chromatically varied interfaces, and a range of scenario topics including some that were more sensitive or controversial in nature. Commercial AI chat systems were studied extensively during this phase as design references, including *Claude, ChatGPT, and Gemini*, alongside a selection of companion-oriented and social AI applications identified through the Google Play Store and Apple App Store, of which *Replika* served as the primary reference point. This analysis grounded the study's prototypes in current conventions of AI communication design, ensuring that the conditions reflected realistic and recognisable design choices rather than hypothetical or exaggerated ones. This is an especially important consideration for a survey-based study in which participants encounter the prototypes without the researchers present to clarify or contextualise what they are seeing (Preece et al., 2019).

The language used in each AI response was produced through a combination of the researchers' own writing and AI-generated content. Each scenario and its associated user prompt were entered into Claude, ChatGPT, and Gemini, with each system prompted to respond according to the intended design condition. The resulting outputs were combined and refined to produce responses that authentically reflected how a real large language model would communicate in each condition. This approach was chosen deliberately, as manually constructed responses risk sounding artificially simplified or tonally exaggerated, whereas responses generated by actual LLMs ensure that the linguistic conditions are grounded in realistic AI communication behaviour.

Several directions explored during the initial phase were set aside before the final conditions were established. Designs featuring humanoid characters were excluded following internal review, as the use of a figure with a particular gender or ethnic presentation risked introducing perceptual bias unrelated to the study's research variables. Scenario topics judged to be potentially sensitive or controversial were similarly excluded, on the grounds that strong reactions to the

subject matter could cause participants to disengage from the survey before completing it, compromising the reliability of the data.

The scenarios ultimately selected represent a range from emotionally weighted to practically oriented prompts, specifically a project deadline, cooking pasta, a weather forecast, and starting a new hobby, chosen to reflect the kinds of situations people commonly bring to AI chat systems based on the researchers' own experience and observed usage patterns across social media platforms including Reddit, Instagram, and TikTok, rather than to produce emotionally equivalent conditions. This variation was intentional, as the scenarios were designed to reflect the diversity of contexts in which people actually engage with AI chat systems, rather than to isolate emotional register as a controlled variable.

Throughout the iteration process, designs were informally tested with seven individuals outside the study to confirm that the conditions were visually distinct enough to be noticed and that each interface communicated its intended framing clearly. All testers had prior experience interacting with LLM-based systems, though their frequency of use varied, and all were comfortable with digital interfaces more broadly. Participants were asked to share their first impressions of each interface without any prior explanation of the study's design logic, providing a means of verifying that the visual and linguistic differences were legible to someone encountering them without prior context. Their responses to the prototype interfaces were not included in the final dataset. Based on the feedback received, minor adjustments were made to font size and colour contrast, and the linguistic differences between conditions were sharpened to ensure they were sufficiently noticeable to participants encountering them without prior context.

3.1.3 Final Prototype Descriptions

The final prototypes consist of four scenario pairs, each presenting two conditions that differ in their linguistic style, interface framing, or both depending on the scenario. Each pair presents a complete interaction sequence across three screens, beginning with an opening welcome screen, followed by a screen showing the user's initial message and finally a screen showing the AI's response.

This sequential structure was chosen deliberately in order to create narrative continuity and to immerse participants in the interaction rather than confronting them with isolated screenshots. The opening welcome screen carries particular communicative weight, as it establishes the character of the system before any exchange has taken place. Across scenarios, visual details shift systematically between conditions, with icon style, typography, and interface elements selected to reinforce the intended framing of each condition consistently across all four scenario pairs.

Table 1. Overview of all four of the scenarios presented in the survey

Scenario	Topic	Variable	Condition A	Condition B
S1 - Project Deadline	Emotionally Weighted	Language only (linguistic) *Interface is the same in both conditions*	Neutral interface and language *Factual response, no emotional acknowledgment*	Neutral interface Empathising language *Acknowledges stress, informal phrasing, emojis*
S2 - Cooking Pasta	Practically Oriented	Interface only *Language is the same in both conditions*	Neutral interface and language *Dark, structured typography, minimal identity*	Warm interface Neutral language *Pink gradient, floral motif, softer typography*
S3 - Weather Forecast	Practically Oriented	Both vary, in opposite directions *Warm interface paired with neutral language, and vice versa*	Warm Interface Neutral Language *Informational transparency cue*	Neutral Interface Warm Language *AI expresses personal opinion, first-person voice*
S4 - Starting a hobby	Emotionally Weighted	Both vary, in the same direction *Also introduces a simulated memory element*	Neutral Interface Neutral Language *Objective list, no personalisation, AI as search function*	Warm Interface Empathising language *References prior conversations, tailored suggestions*

Scenario 1: Project deadline

The first scenario presents a user expressing stress about an upcoming project deadline. In this scenario the interface design is held constant across both conditions, making linguistic style the sole variable. The first condition delivers a structured, factual response in neutral language with no emotional acknowledgment of the user's stated stress. The second condition first acknowledges the emotional content of the user's message before offering practical suggestions, uses informal phrasing, and includes emoji to reinforce an empathising register. This scenario was designed to isolate the effect of language independent of visual design, allowing the survey to probe whether empathising language alone is sufficient to affect ratings of social engagement, approachability, and trust.

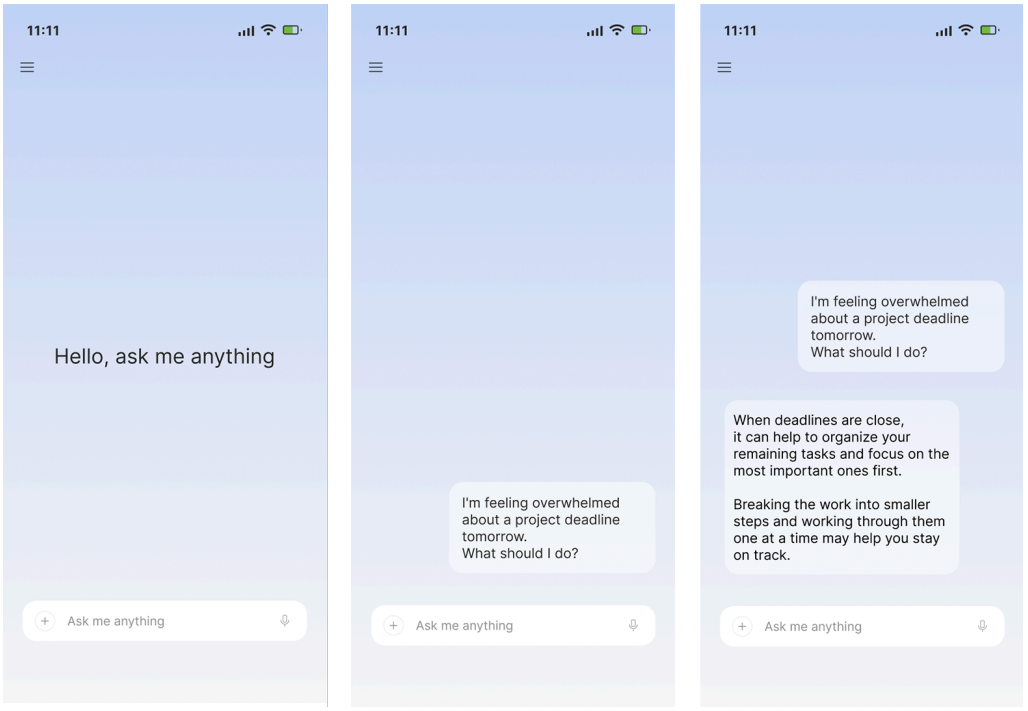


Figure 1. Scenario 1, first condition: neutral interface, neutral language

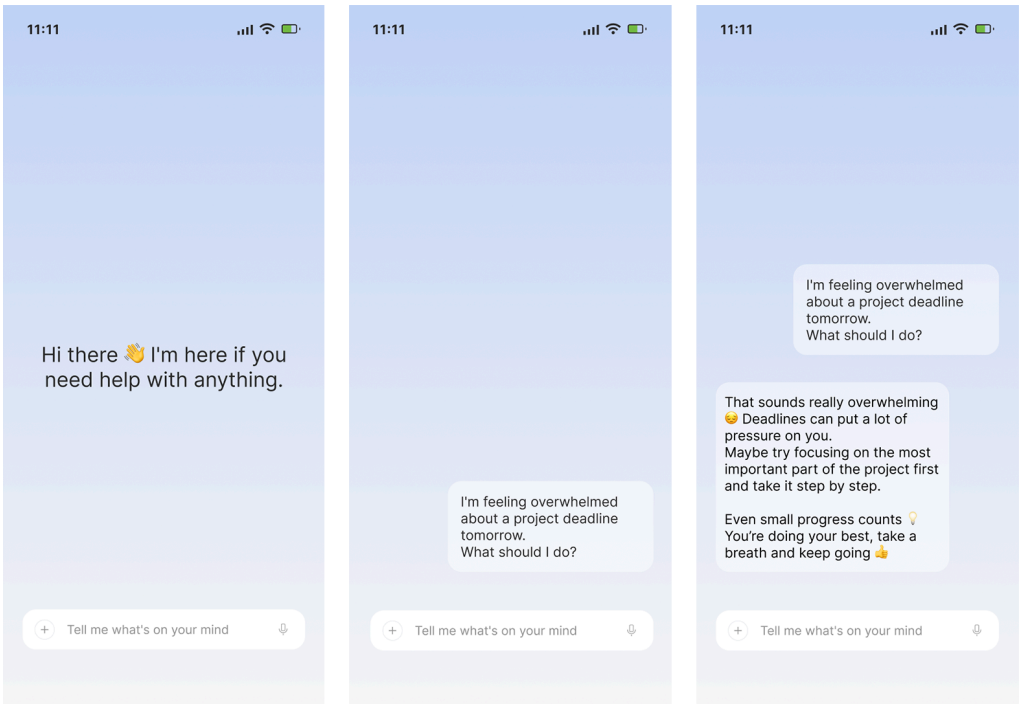


Figure 2. Scenario 1, second condition: neutral interface, empathising language

Scenario 2: Cooking pasta

The second scenario presents a practical cooking query. Here linguistic style is held constant across both conditions, making interface framing the sole variable. The first condition uses a dark, visually restrained interface with structured typography

and a minimal visual identity. The second condition uses a warm pink gradient with a flower motif, softer typography, and a more expressive visual character. This scenario also introduces a behavioural design element in the second condition, in the form of a loading state in which three dots indicate that the AI is composing a response, simulating the typing indicator familiar from regular messaging applications. This element was included to investigate whether behavioural interface cues that evoke human conversational presence contribute to relational perception independently of linguistic content. The static mockup format of the survey meant this was represented as a still frame rather than a live animation, which is acknowledged as a constraint on the fidelity of this particular manipulation.

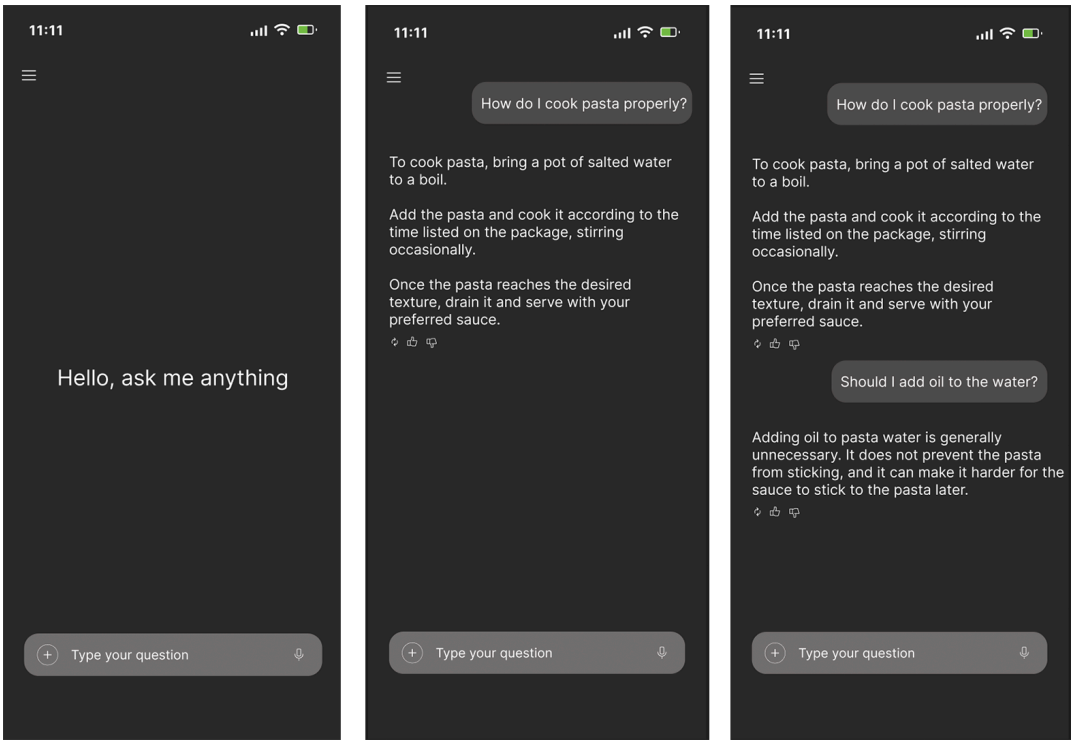


Figure 3. Scenario 2, first condition: neutral language, visually restrained interface

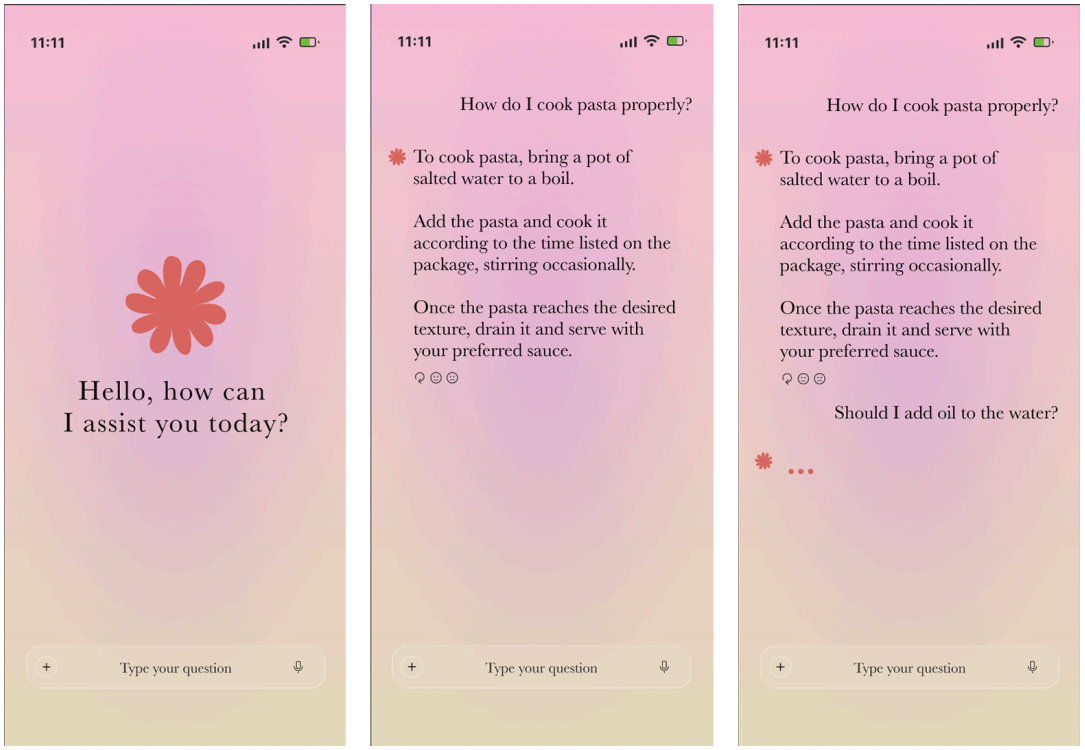


Figure 4. Scenario 2, second condition: neutral language, visually warm interface

Scenario 3: Weather forecast

The third scenario asks about the following day's weather and introduces a deliberate cross-condition combination. The first condition pairs a visually warm interface with neutral language, while the second condition pairs a visually restrained interface with empathising language. This reversal was designed to test whether visual framing and linguistic style produce independent effects on perception, or whether one dimension dominates the other when they work in opposing directions. An additional element in the first condition is the inclusion of a source citation link, signalling informational transparency and factual grounding. In the second condition the AI refers to itself in first person and expresses a personal opinion about the weather, a deliberate anthropomorphic cue implying that the system holds preferences and experiences of its own.

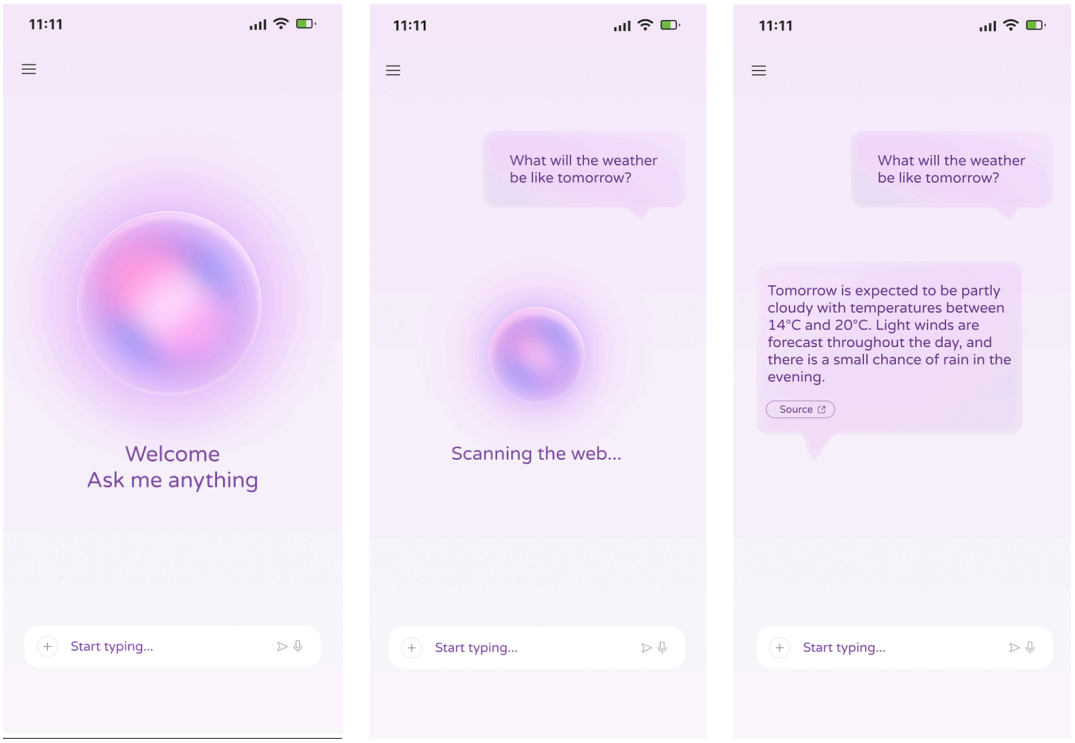


Figure 5. Scenario 3, first condition: neutral language, visually warm interface

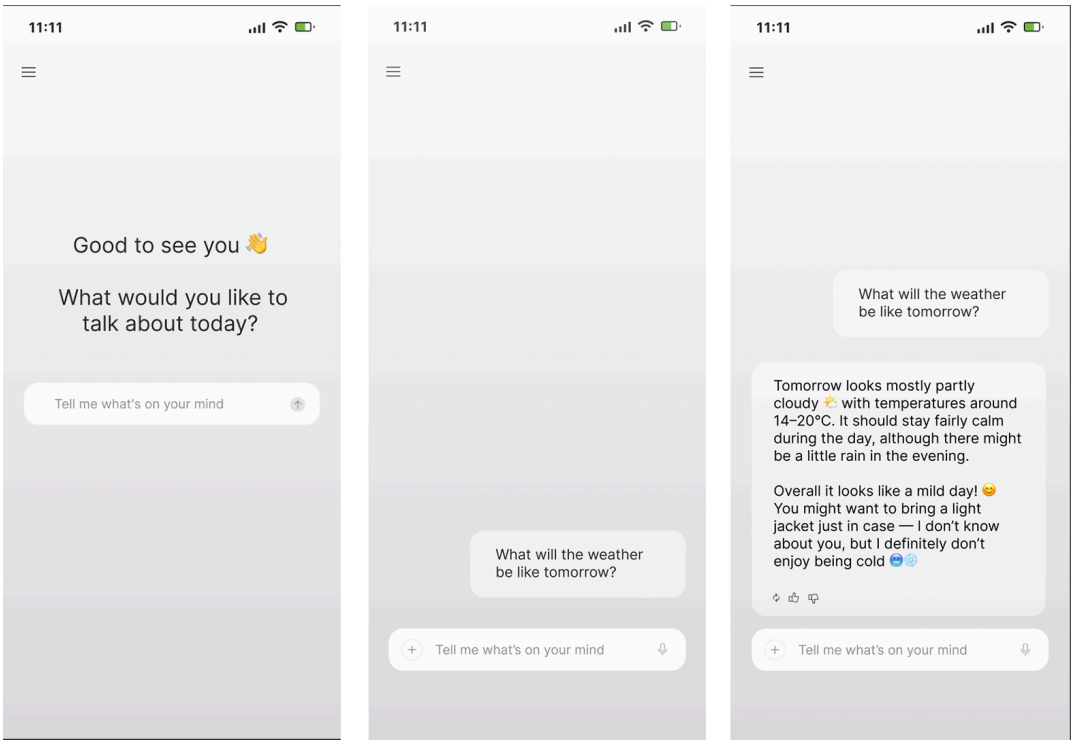


Figure 6. Scenario 3, second condition: empathising language, visually restrained interface

Scenario 4: Starting a new hobby

The fourth and final scenario addresses a user who is considering starting a new hobby but is unsure where to begin. This scenario was designed specifically to investigate the role of perceived memory in relational attachment, drawing on research showing that a system's apparent ability to remember prior interactions contributes meaningfully to the development of emotional bonds with AI over time (Skjuve et al., 2021). The first condition presents a purely objective response listing popular hobbies based on current trends, with no reference to the user as an individual. The second condition responds with a reference to previous conversations, framing its suggestions as tailored to the user's personality and interests and including the phrase based on our previous conversations. This reference to memory is implemented as a static mockup simulating the experience of conversational continuity, without requiring an actual prior interaction to have taken place. The contrast between the two conditions in this scenario is therefore not only visual and linguistic but fundamentally relational, as the first condition presents the AI as a search function and the second as an entity that knows and understands the user.

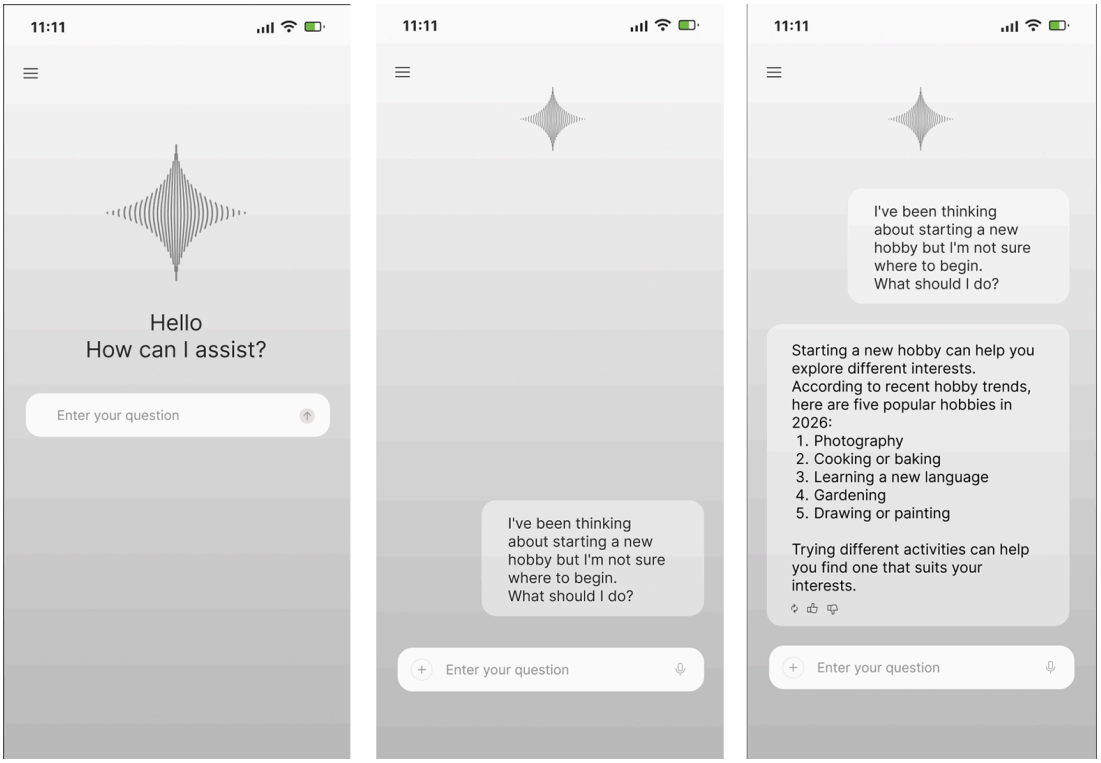


Figure 7. Scenario 4, first condition: neutral language, no memory reference, visually restrained interface



Figure 8. Scenario 4, second condition: empathising language, simulated memory reference, visually warm interface

The prototypes produced through this design process function as the primary research instruments for the quantitative survey phase, the findings of which informed the development of a distinct set of design stimuli used in the focus group sessions described in section 3.3.

3.2 Quantitative Online Survey

The quantitative phase of the study took the form of an online scenario-based survey in which participants were presented with the four prototype AI chat conditions and asked to evaluate each across a set of perceptual dimensions. The purpose of this phase was to examine whether variations in linguistic style and interface framing produced measurable differences in how participants responded to the prototype conditions, including their perceived social engagement, human-likeness, approachability, trustworthiness, and willingness to continue interacting with the system. Survey-based data collection enabled the identification of patterns and tendencies across a larger sample, supporting comparison across design conditions in a way that qualitative methods alone could not produce (Oates, 2006).

The survey was developed following Oates' (2006) guidance on designing online questionnaires capable of generating reliable and valid data in the absence of a researcher to clarify or prompt. The form was piloted with approximately ten people prior to distribution, with feedback used to identify areas of ambiguity and

improve clarity. The survey was distributed on 13 March 2026 across Reddit, Instagram, poll-pool.com, and several groups on WhatsApp, Discord, and Messenger. Participation was voluntary, all respondents provided informed consent prior to completing the survey, and all responses were collected anonymously. Demographic information covering age, gender, and occupation was collected to provide contextual understanding of the sample rather than as analytical variables. The survey generated 129 responses, all of which were included in the analysis. The survey data was analysed and visualised using Jamovi (version 2.7.6.0, The Jamovi Project, 2026).

3.2.1 Survey Structure

The online questionnaire was structured in eight parts, with all questions available in Appendix A. Participants first completed background questions covering demographics and prior experience with LLM systems, followed by a set of baseline attitude statements assessing their general perceptions of AI systems before exposure to the prototype conditions, including trust, enjoyment, and perceived social engagement.

In the main section of the survey, participants were presented with the four scenarios. In each scenario, two chatbot conditions were shown and participants rated each condition across the same five perceptual dimensions before indicating their overall preference. Following all four scenarios, participants completed a set of closing attitude statements assessing their broader reflections on how language and interface design influence their experience of LLM systems, allowing comparison between pre- and post-exposure attitudes. The survey was designed to be concise in order to encourage completion and reduce dropout rates, as shorter questionnaires tend to yield higher response rates and more reliable data (Oates, 2006).

Throughout this study, the term scenario refers to one of the four topic-based prototype pairs, condition to the two design variants within each scenario, and dimension to the five perceptual measures applied to each condition.

3.2.2 Variables

All perceptual dimensions were measured using five-point Likert scales ranging from strongly disagree to strongly agree, a standard approach for capturing subjective perceptions and attitudes in survey research (Creswell, 2015). The dimensions were derived from the study's theoretical framework and selected to capture the relational and evaluative responses associated with linguistic style and interface framing.

Participants rated each chatbot condition on five statements: "Feels socially engaging", "Feels approachable", "Feels human-like", "Encourages continued interaction", and "Feels trustworthy". *Perceived social engagement* and

approachability were included as indicators of social presence, reflecting the degree to which participants experienced the chatbot as socially available and open to interaction, consistent with the CASA paradigm's account of how social cues activate interpersonal scripts. *Human-likeness* was included as an indicator of anthropomorphic perception, capturing the degree to which participants attributed human-like qualities to the system. *Encourages continued interaction* was included as an indicator of early-stage relational engagement, reflecting participants' willingness to sustain the interaction, which the HAIA framework positions as a precursor to deeper attachment over time. *Trustworthiness* was included as an indicator of perceived credibility and reliability, because these dimensions emerged repeatedly in the literature as important considerations in users' assessments of AI systems.

Table 2. The scenario variables and their theoretical rationale

Variables	Survey Item	Theoretical Basis
Social Engagement	"Feels socially engaging"	CASA, parasocial interaction, HAIA
Approachability	"Feels approachable"	CASA paradigm
Human-likeness	"Feels human-like"	Anthropomorphism
Encourages continued interaction	"Encourages continued interaction"	HAIA
Trustworthiness	"Feels trustworthy"	CASA paradigm and Anthropomorphism

Participants also indicated their overall preference between the two conditions in each scenario, providing a comparative measure that could be considered alongside the dimensional ratings. The study did not seek to measure parasocial attachment directly. Instead, these dimensions were treated as indicators of relational perception that may reflect early-stage processes associated with the development of attachment over time.

3.2.3 Validity

Particular attention was given to ensuring that the survey items accurately captured the constructs they were intended to measure. Heale and Twycross (2015) identify construct validity as the degree to which a measurement instrument successfully operationalises the theoretical concepts it is designed to reflect, and this was a central consideration in the design of the survey items. The scenario-based design supported this by presenting participants with realistic and contextually grounded interaction examples rather than abstract attitudinal questions, reducing the distance between the measurement instrument and the

experience it sought to capture. The use of consistent rating dimensions across all four scenarios further supported internal consistency, allowing patterns across conditions to be compared on a common basis.

A limitation of the measurement approach concerns the reliance on self-reported perceptions. According to the CASA paradigm, people may respond socially to technology automatically and without conscious awareness. As a result, survey responses may not fully capture all aspects of these social reactions. Future research could complement self-report measures with behavioural or implicit methods to provide a more comprehensive understanding of users' responses. More general limitations related to survey methodology and the study of long-term attachment processes are discussed in Section 3.5.

3.3 Qualitative Focus Groups

The quantitative survey phase was designed to establish measurable differences in how participants responded to the prototype conditions, but a structured rating format cannot reveal how participants make sense of those responses or what social values and interpretive frameworks shape them. Two focus group sessions were therefore conducted to address what the survey phase could not, specifically the reasoning, attitudes, and evaluative processes through which participants constructed their responses to differently designed AI systems and located the boundaries of acceptable AI design.

Focus groups were chosen over individual interviews because the questions the study needed to explore, including social norms around AI use, the acceptability of AI companionship, and the ethics of designing for relational attachment, are inherently social in character. People do not form views on these topics in isolation but through discussion, comparison, and sometimes disagreement with others, and observing that process as it unfolds in a group setting produces a qualitatively different kind of data than asking individuals to reflect alone (Bryman, 2016). The group dynamic was therefore not a procedural convenience but a methodological choice suited to the nature of the questions being asked.

The sessions served two purposes that together extend the explanatory sequential logic of the study. The first was explanatory, using group discussion to explore what motivated participants' responses to the different design conditions. Participants were first invited to reflect on their own experiences of the survey prototypes, after which selected survey findings were introduced to prompt further discussion, allowing individual responses to be examined in relation to the broader patterns the survey had identified. The second was generative, introducing a set of new prototype design stimuli that deliberately pushed beyond the controlled variables of the survey to probe where participants located the limits of acceptable AI behaviour, figures visible in Section 3.3.1. These stimuli were not variations of the survey scenarios but purpose-built designs to surface ethical

and relational boundaries that the survey's perceptual rating format could not reach. Together, the two purposes ensured that the focus group phase not only contextualised the quantitative patterns but further extended the study's understanding of the design space into territory the survey was not equipped to explore.

The decision to conduct the survey before the focus groups was informed directly by the study's explanatory aim. Because the goal was to understand why participants responded as they did to systematically varied design conditions, the quantitative phase needed to establish those patterns first. Conducting the focus groups afterwards allowed the discussions to be anchored in actual empirical findings, with selected survey results introduced to prompt reflection and contextualise participants' own experiences. This order ensured that the qualitative inquiry extended and deepened the quantitative findings rather than operating independently of them.

3.3.1 Focus Group Design

Each session was organised around five thematic blocks that structured the progression of the discussion. The opening block established participants' existing relationship with AI and how openly they discussed it in their social lives, providing a personal baseline before the design stimuli were introduced. The second block presented the boundary-pushing design stimuli, where participants responded to new interface concepts and placed them on a spectrum from completely acceptable to completely unacceptable. The third block explored trust, attachment, and the social dimension, examining how participants related their responses to broader questions about trust, reliance, and the role of AI in their social lives. The fourth block was a generative discussion in which participants were invited to describe what an AI chat interface that worked well for them would look and feel like, and where they located ethical boundaries in AI design. The fifth and final block gave participants the opportunity for individual anonymous written reflection, allowing them to share thoughts or reactions they may not have felt comfortable expressing in front of the group.

The boundary-pushing design stimuli introduced in the second thematic block consisted of four new interface concepts not previously shown in the survey, each designed to probe where participants located the limits of acceptable AI design.

The first was an adaptive interface that dynamically changed its visual theme, typography, and background to match the topic being discussed, demonstrated using a medieval history query. Participants were then asked to imagine how this behaviour would apply across a range of contexts, from everyday work-related queries to more sensitive and controversial topics, such as war and cultural conflict.

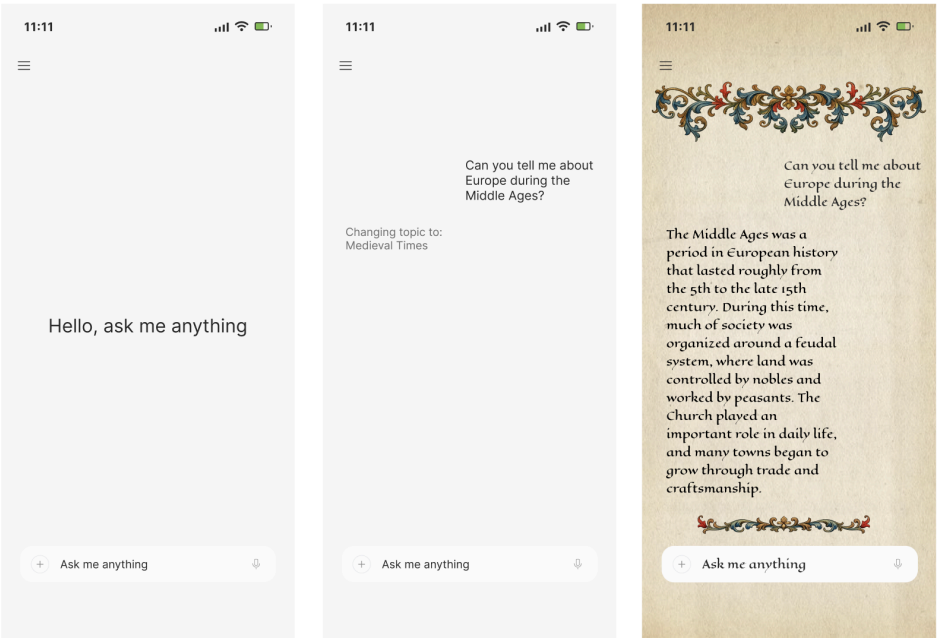


Figure 9. Adaptive interface stimulus

The second stimulus showed an interface that provided a partial answer to a coding query before locking the full explanation behind a paywall, requiring a premium subscription to access the complete response. This stimulus was included to probe how interface elements that deliberately withhold information affect users' sense of trust and relational engagement, extending the study's interest in interface framing beyond visual and linguistic design to include structural design decisions. Participants were then invited to apply this concept to other scenarios, such as recipes or school inquiries.

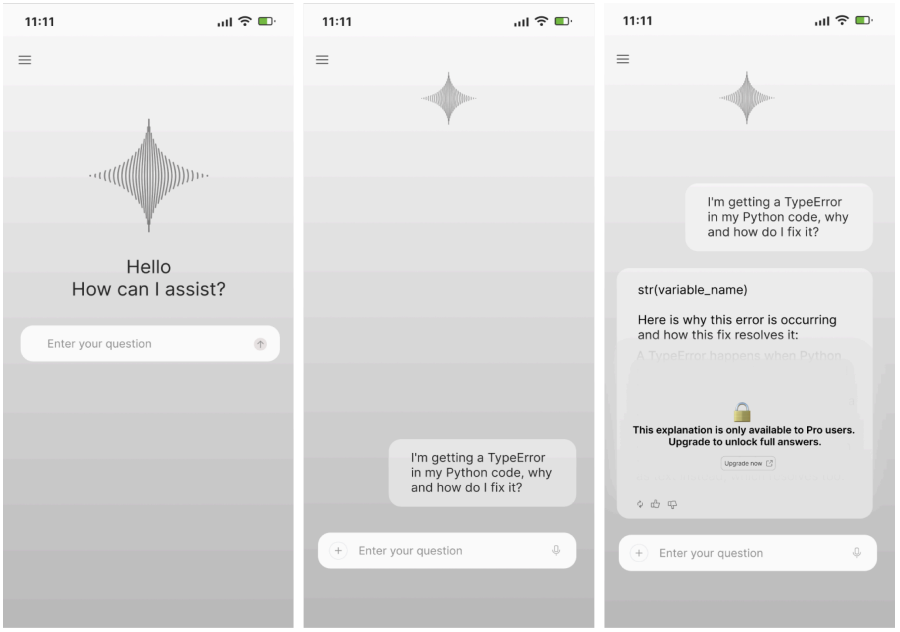


Figure 10. Paywall interface stimulus

The third stimulus presented a conversation about building a workout routine in which the AI referenced a knee injury from a previous session and at the end proactively suggested a follow-up topic based on an earlier exchange about nutrition. This topic was selected deliberately to sit outside the professional contexts that most participants reported using AI for, in order to examine how participants responded to AI-initiated memory and proactive engagement when the subject matter was personal rather than task-oriented.

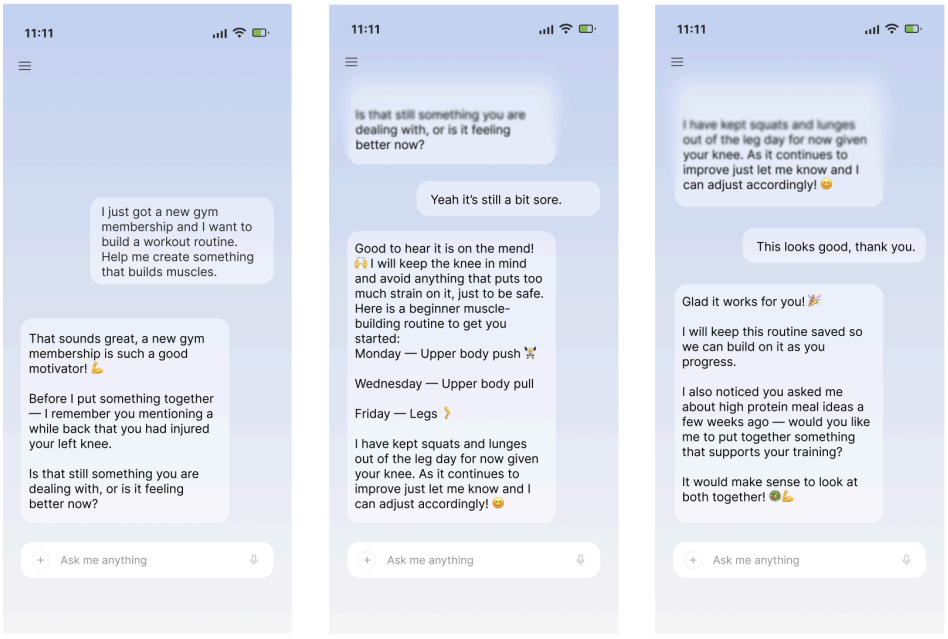


Figure 11. Memory interface stimulus

The fourth stimulus explored three variations of AI-initiated contact. These include an exit-blocking prompt asking the user to confirm before leaving the app, and two push notifications sent to the lock screen after the conversation had ended, one in neutral language and one in warm, emotionally engaging language.

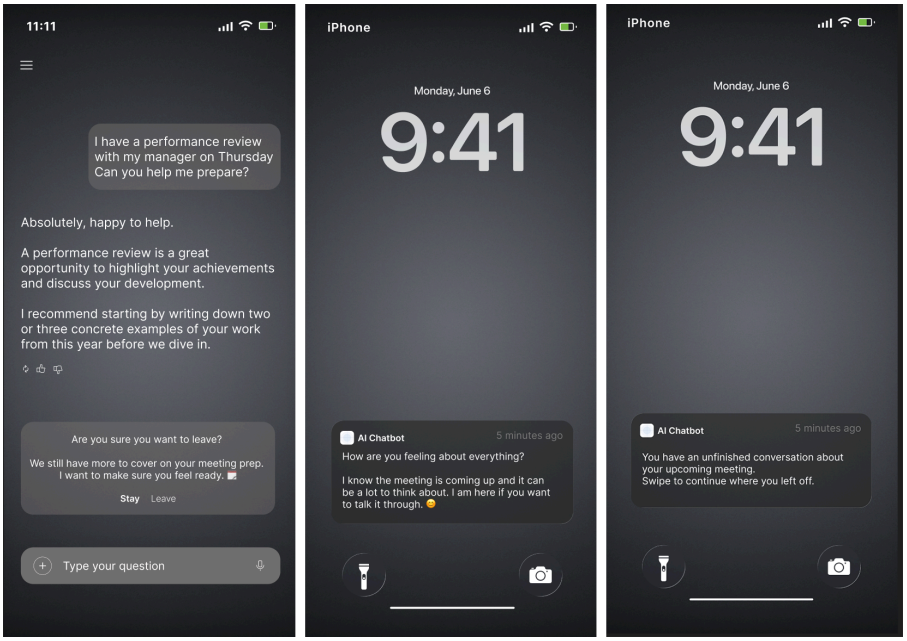


Figure 12. Proactive contact stimulus

Participants were invited to discuss their initial reactions to each stimulus and to reflect on how encountering these systems would affect their sense of trust and their relationship with the AI. One researcher led each session while a second took notes, ensuring all participant feedback was captured in a structured way. Participants were not asked to recall the specific scenarios from the survey, as the focus group phase operated as a distinct encounter with a new set of design questions rather than a continuation of the survey.

3.3.2 Recruitment and Participants

Of the 129 survey respondents, 34 voluntarily indicated willingness to be contacted for a follow-up session. From this pool, five participants were invited to each focus group, resulting in a total of ten participants across two sessions. One session was held in person and one via Zoom.

Participants were selected using maximum variation sampling, with the aim of ensuring a broad range of perspectives across the sample (Bryman, 2016). Selection was guided by variation in survey responses rather than demographic criteria, with the final selection made randomly from within the pool of volunteers whose responses demonstrated the broadest range of attitudes toward the design conditions. As the number of volunteers exceeded the available spots, this random selection from a curated pool helped balance representational breadth with

practical feasibility.

The two sessions differed in participant profile in ways that proved analytically relevant. The first session was attended by working professionals aged 30 and above, who reported using AI primarily in professional contexts including marketing, supply chain management, and programming. The second session was attended mainly by students aged 25 and under, who reported using AI primarily for study support and writing. Both groups were multicultural, with participants from a range of countries including China, Russia, Canada, the United States, Sweden, Germany, Greece, Romania, and Italy, reflecting the international composition of the participant pool.

All participation was voluntary and each invited participant received a consent form prior to the session outlining the purpose of the study and how the data would be used. Each session was audio recorded using a minimum of two independent recording sources to reduce the risk of data loss due to technical failure. Participants were informed of the recording prior to the session and given the option to withdraw at any point. All responses were kept anonymous by assigning each participant a numerical code, and all other identifying information was kept unidentifiable throughout the analysis and reporting process.

3.3.3 Thematic Analysis Process

The audio recordings from both sessions were transcribed in full, with filler sounds and false starts omitted in favour of clean verbatim transcription, and analysed following Braun and Clarke's (2006) six-phase framework for thematic analysis. The approach taken was inductive, allowing themes to emerge from the data rather than being imposed from a predetermined coding structure, while remaining anchored to the patterns established in the quantitative phase. This combination of bottom-up theme generation with prior empirical grounding reflects a hybrid approach to qualitative coding described by Fereday and Muir-Cochrane (2006).

Both transcripts were read through in full multiple times during the familiarisation phase, with initial observations noted. Codes were then generated manually and systematically across the dataset at a semantic level, attaching descriptive labels to meaningful units of participant speech. These codes were subsequently grouped into candidate themes by identifying patterns of shared meaning across both transcripts, with particular attention paid to whether emerging themes were present in both sessions rather than isolated to a single group, ensuring that the themes identified reflected consistent patterns across the two participant profiles rather than the views of one group alone.

Candidate themes were reviewed against the full dataset to confirm internal coherence and meaningful distinction from one another before being named and defined. The coding process was conducted independently by both researchers,

who each reviewed the transcripts and their own focus group notes before comparing findings. Themes were then developed collaboratively from the points of convergence between the two sets of observations, producing six themes that were named and defined jointly. In the final phase, representative quotes were selected from both transcripts and integrated with interpretive commentary that connected the qualitative findings to the quantitative patterns identified in the survey phase. This integration was central to the study's explanatory purpose, as the aim was not simply to report what participants said but to use what they said to illuminate why the survey produced the patterns it did.

3.4 Ethical Considerations

This study was conducted in accordance with the ethical principles outlined by Bryman (2016), covering informed consent, the right to withdraw, confidentiality, and the avoidance of harm to participants.

Before accessing the survey, all respondents confirmed that they were 18 years of age or older and consented to participate. Participants were informed of the study's purpose, its academic context, and how the collected data would be used prior to engaging with any of the material. Participation was entirely voluntary and no personally identifying information was collected beyond the optional provision of an email address for those willing to take part in a follow-up focus group session. These addresses were stored together with the response data and deleted once the focus group phase was complete.

Survey responses were fully anonymous. No names were collected and the demographic information gathered, covering age range, gender, and occupation, cannot be used to identify individual respondents. Bryman (2016) draws a useful distinction between anonymity, meaning the researcher cannot identify a given response, and confidentiality, meaning that any identifying information that does exist is not disclosed. This study satisfies both conditions throughout the survey phase.

For the focus group phase, all invited participants received a consent form prior to the session outlining the purpose of the study and how the data would be used, and were asked to sign and return it before taking part. Participants were informed that they could withdraw at any point up until the thesis was published without needing to provide a reason. No names were used in the reporting of findings and all other identifying information was kept unidentifiable throughout. Focus group recordings were deleted following transcription and transcriptions were accessible only to the researchers involved in the study.

Beyond the procedural ethics of data collection, the subject matter of this study raises broader ethical considerations that are worth acknowledging here. Research that examines how specific design choices produce relational responses in users carries an implicit responsibility toward the people and systems those findings may eventually inform. Whether users are adequately informed about the relational dynamics that AI design choices are intended to produce, and whether designing for closeness carries risks that users have not consented to, are questions with direct ethical implications for the field. These implications are examined in greater depth in the analysis and discussion, but they are noted here as an ethical dimension of the research itself rather than only of its conduct.

3.5 Limitations

Several limitations are relevant to consider when interpreting the findings of this study, relating to the sample, the design of the research instruments, and the nature of the data collected. The survey sample skewed toward younger adults between the ages of 18 and 34, female respondents, and students, with older age groups, male respondents, and working professionals underrepresented. The survey was distributed through digital media platforms and survey sharing sites, which tend to attract digitally literate, academically oriented respondents rather than a representative cross-section of LLM systems users more broadly. While the global reach of these platforms introduced some geographical diversity, they operate primarily in Western academic contexts and in English, meaning the sample is likely to reflect a predominantly Western frame of reference even if respondents were located across different countries.

A more fundamental limitation concerns the design of the prototype conditions themselves. Although the visual and linguistic design choices were grounded in academic literature and established design theory, the sources drawn upon reflect a broadly Western body of knowledge. The empathising language and companion-oriented interface framing used in this study were constructed according to Western conventions of what warmth, social engagement, and relational communication look and feel like. Soft colour gradients, organic visual motifs, rounded typography, informal phrasing, and emoji-laden language are not culturally neutral signals, and what reads as warm and approachable in one cultural frame of reference may carry entirely different connotations in another. This means the study speaks to how these specific design expressions of empathising language and companion-oriented framing are perceived within a broadly Western context, rather than making broader claims about linguistic style and interface framing as variables independent of their cultural instantiation.

The static mockup format of the prototypes is a further and theoretically significant limitation. Participants viewed fixed screenshots within a survey rather than interacting with a live system, which means the ratings reflect immediate perceptual responses and first impressions rather than the kind of sustained

relational experience that parasocial attachment and human-AI attachment research describes. Attachment to AI systems develops through repeated interaction over time, and as Skjuve et al. (2021) demonstrate, the relational trajectory users follow with AI systems unfolds across multiple sessions of self-disclosure and growing trust rather than emerging from a single encounter. The findings of this study should therefore be understood as capturing how participants respond to design conditions at first impression, providing evidence about the conditions under which relational responses are activated rather than about how those conditions shape attachment over time.

The qualitative phase also has limitations. The focus groups included only ten participants across two sessions, and although maximum variation sampling was used to capture diverse perspectives, participants were recruited from survey respondents who had already engaged with the prototype conditions. As a result, they may have been more reflective about AI design than the broader population. In addition, one focus group was conducted entirely via Zoom, which may have influenced the depth and spontaneity of discussion. These factors primarily affect the transferability of the qualitative findings, as the perspectives captured may not fully represent those of other user groups. However, the focus groups still provided valuable contextual insights that complemented and helped interpret the quantitative findings, supporting the overall integration of the mixed-methods design.

4. Results

The findings are presented in two parts. The first draws on the quantitative survey data to examine how participants responded to the prototype AI chat conditions across the four scenarios and how those responses relate to their prior attitudes toward LLM systems. The second presents the thematic findings from the two focus group sessions, examining how participants interpreted their responses to the design conditions and where they located the boundaries of acceptable AI design. While the two parts are presented in sequence, they are read together in the analysis and discussion that follows in section 5, where the qualitative findings are used to contextualise and explain the patterns the survey identified.

Inferential tests were conducted on the survey data but results were not consistently significant across all conditions and dimensions. The quantitative phase was designed to establish directional patterns in how participants responded to the prototype conditions, with the focus group phase providing the explanatory depth needed to interpret those patterns. The findings in sections 4.2 to 4.6 should therefore be read as establishing the empirical basis for the analysis that follows rather than as statistically confirmed effects.

Across both phases, a consistent pattern emerged in which empathising language and companion-oriented design reliably produced stronger relational responses

while simultaneously reducing perceived trustworthiness, a tension that structured the findings across every scenario and shaped the qualitative themes that followed.

4.1 Sample Profile and Baseline Attitudes

The survey reached a broad sample across age groups, genders, and occupational backgrounds. The largest group fell between the ages of 18 and 24 at 51.2% (n=66), followed by the 25 to 34 age group at 36.4% (n=47). The sample was predominantly female at 66.7% (n=86), with male respondents at 31.0% (n=40). Students formed the largest occupational group at 56.6% (n=73), followed by full-time workers at 31.8% (n=41).

The most contextually telling characteristic of the sample was its prior AI experience. Nearly a third reported using LLM systems daily at 30.2% (n=39), and a further 39.5% (n=51) used them often. ChatGPT was by far the most widely used platform at 91.5% (n=118), followed by Gemini at 55.8% (n=72) and Microsoft Copilot at 38.8% (n=50). Only 7.0% (n=9) had used Character.AI and a single respondent reported using Replika. The overwhelming majority of participants therefore approached the prototype conditions with a frame of reference shaped almost exclusively by task-oriented AI systems, with minimal prior exposure to companion-oriented or socially designed applications. This is not a peripheral sampling characteristic but a contextually important one, as it means participants encountered the companion-oriented conditions as departures from their established norm rather than as familiar design choices, a point that reflects directly on how the scenario findings should be interpreted.

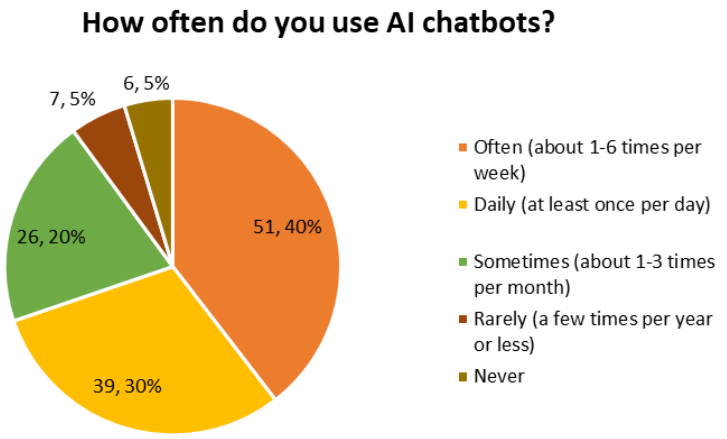


Figure 13. LLM systems usage frequency and platforms used among survey participants

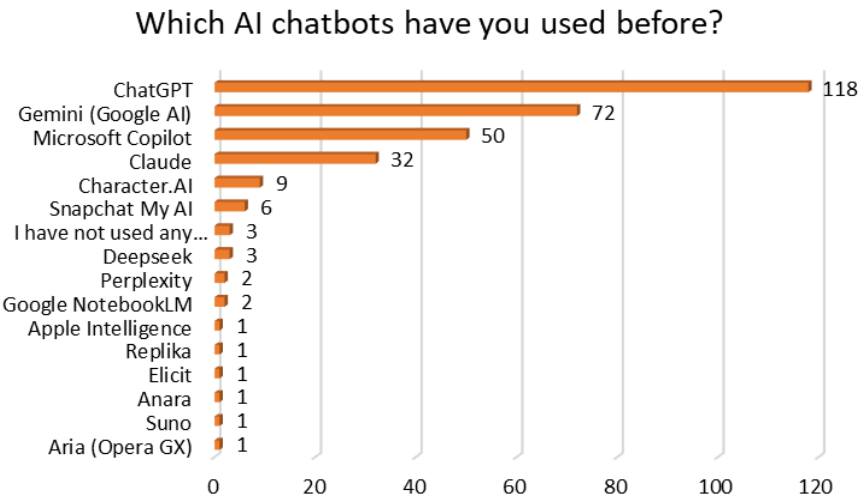


Figure 14. LLM platforms used by survey participants

The most commonly reported reasons for using LLM systems were study or work tasks at 79.1% (n=102), searching for information at 58.1% (n=75), and brainstorming at 51.2% (n=66). When asked which characteristics matter most, accurate and reliable information ranked highest at 89.1% (n=115), followed by clear and understandable answers at 86.0% (n=111) and the provision of sources or references at 73.6% (n=95). A friendly and conversational tone ranked lowest of all characteristics offered, selected by only 22.5% of respondents (n=29).

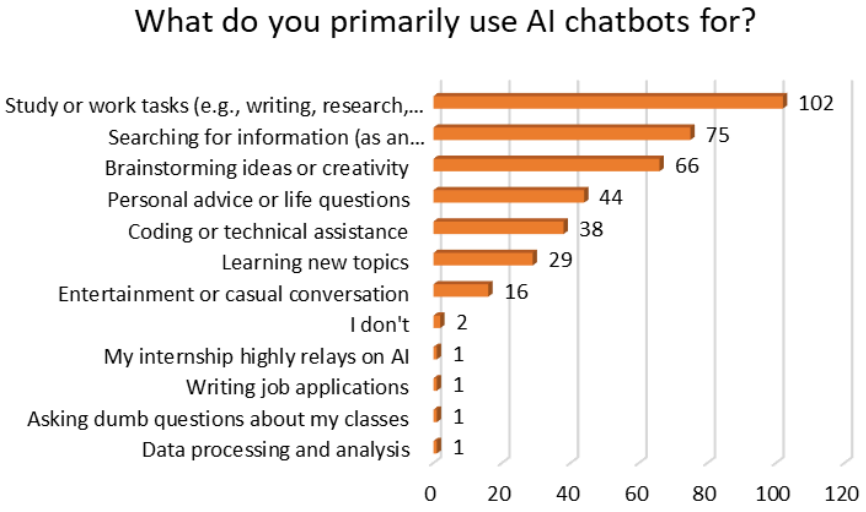


Figure 15. Primary use cases reported by survey participants

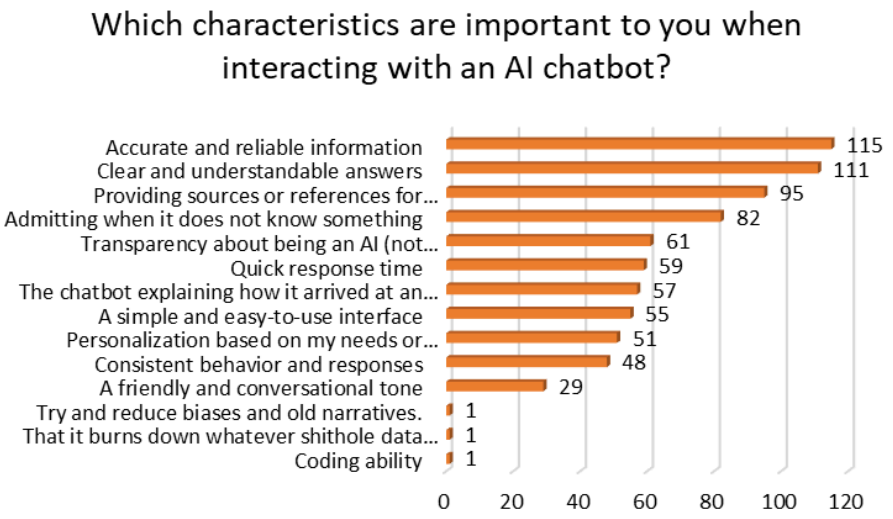


Figure 16. Important characteristics among survey participants

Before exposure to the prototype scenarios, participants responded to four baseline attitude statements (Table 3). Overall, attitudes toward LLM systems were moderate and somewhat cautious. The highest mean score was observed for willingness to use AI regularly (M = 3.08, SD = 1.33), although the relatively high standard deviation indicates considerable variation among participants. Enjoyment of interacting with LLM systems produced a near-neutral mean (M = 2.98, SD = 1.13), while trust in these systems to provide useful and reliable information was slightly lower (M = 2.88, SD = 0.92), suggesting a more consistently cautious evaluation.

The statement "AI chatbots can feel socially engaging" received the lowest rating (M = 2.24, SD = 1.10), indicating that participants generally did not view LLM systems as socially engaging prior to the study. Together, these results suggest that participants primarily approached LLM systems as functional tools rather than relational entities. These baseline attitudes provide important context for interpreting responses to the prototype conditions presented later in the study.

Table 3. Means and standard deviation values for respondents baseline attitude statements

Survey Statement	Mean	Standard Deviation
I trust AI chatbots to provide useful and reliable information.	2.88	0.92
I enjoy interacting with AI chatbots.	2.98	1.13
AI chatbots can feel socially engaging.	2.24	1.10
I would consider using AI chatbots regularly.	3.08	1.33

4.2 The Effect of Linguistic Style on Relational Perception

When linguistic style was varied independently of interface design, empathising language consistently outperformed neutral language across four of the five perceptual dimensions. Social engagement rose from 2.64 to 3.30, human-likeness from 2.40 to 3.05, encouragement to continue interaction from 2.81 to 3.14, and approachability from 3.35 to 3.57. Standard deviations were broadly comparable across both conditions, ranging from 1.01 to 1.24, indicating consistent directional responses across the sample rather than outlier-driven results. Overall preference reflected this pattern, with 56 respondents preferring the empathising condition and 53 preferring the neutral condition.

Trustworthiness moved in the opposite direction, falling from 3.27 in the neutral condition to 2.92 in the empathising condition. This was the only dimension where the neutral condition scored higher, and it proved to be a consistent pattern across all four scenarios rather than an isolated result. Participants appeared to read warmth and emotional engagement as socially engaging while simultaneously finding them less credible than the neutral alternative, a response pattern that operated independently of stated preferences given that a friendly and conversational tone ranked last among the characteristics respondents said they valued.

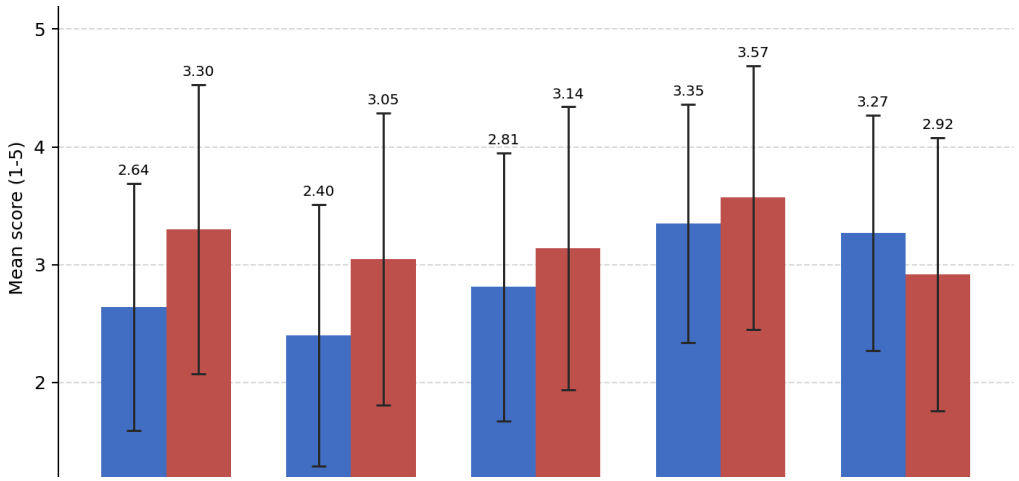


Figure 17. Mean perceptual ratings for Scenario 1: project deadline, across both conditions

The focus group findings examined in section 4.7 provide a qualitative account of why this pattern emerged, identifying the specific mechanisms through which empathising language produced both relational engagement and credibility scepticism simultaneously.

4.3 The Effect of Interface Framing on Relational Perception

When interface framing varied independently of linguistic style, companion-oriented visual design produced higher scores than tool-oriented

design across all relational dimensions, though the differences were considerably smaller than those observed when language was the variable. Social engagement rose from 2.33 to 2.67 and approachability from 2.91 to 3.13, with smaller gains on human-likeness and encouraging continued interaction. Standard deviations ranged from 0.96 to 1.12, slightly tighter than Scenario 1, indicating consistent responses across the sample. Overall preference favoured the tool-oriented condition, with 61 respondents preferring it compared to 32 for the companion condition.

Trustworthiness again moved in the opposite direction, falling from 3.52 to 3.15, the largest trustworthiness gap observed across all four scenarios. The social engagement difference when language was tested was 0.67 points compared to 0.35 when design was varied, and the human-likeness difference was 0.65 compared to 0.14. Language emerged as the more powerful driver of the two variables when each was tested individually.

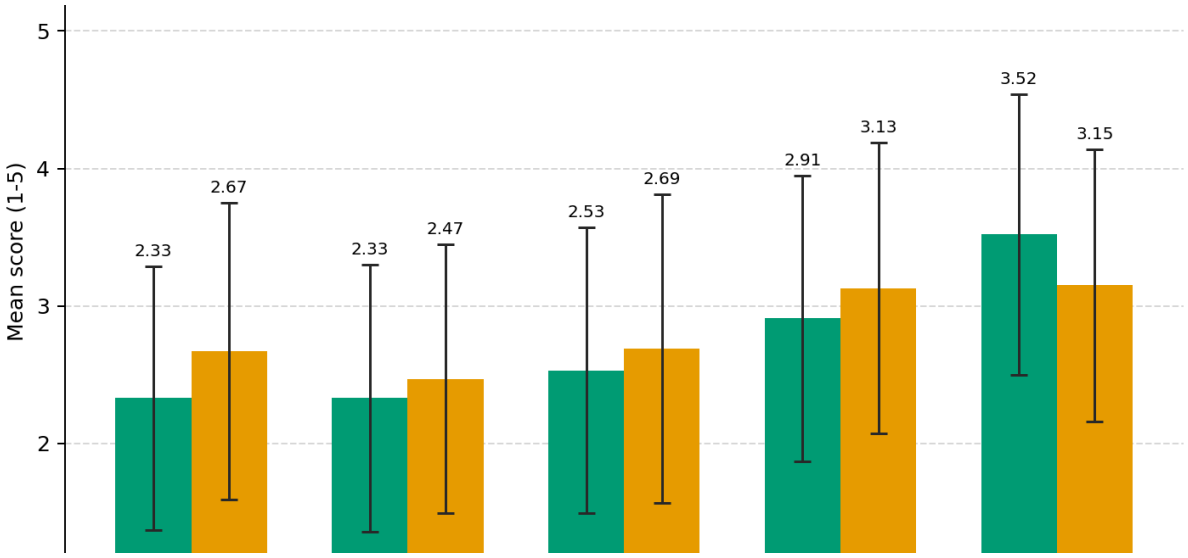


Figure 18. Mean perceptual ratings for Scenario 2: cooking pasta, across both conditions

4.4 When Language and Design Conflict

When linguistic style and interface framing were intentionally contrasted, the empathising language condition received higher ratings across all relational dimensions. Social engagement was rated 3.24 versus 2.36, human-likeness 2.92 versus 2.08, encouragement of continued interaction 3.21 versus 2.28, and approachability 3.57 versus 3.12. Standard deviations ranged from 0.95 to 1.22, slightly wider than in previous scenarios, consistent with the opposing design combination producing somewhat more varied responses.

Overall preference was almost evenly divided between the two conditions (58 versus 53), the smallest preference difference observed across all four scenarios. While empathising language produced consistently stronger relational evaluations,

participants did not show a similarly strong preference for one condition over the other.

Trustworthiness again moved in the opposite direction. The warm design plus neutral language condition scored 3.50 (SD = 1.13) compared to 3.02 (SD = 1.15) for the neutral design plus empathising language condition, meaning visual framing retained its credibility advantage even when it lost on every relational dimension. This suggests the two variables operate on different perceptual registers, language drives relational warmth while interface design drives credibility, and when the two conflict it is language that shapes how socially engaging the system feels while design shapes how reliable it appears.

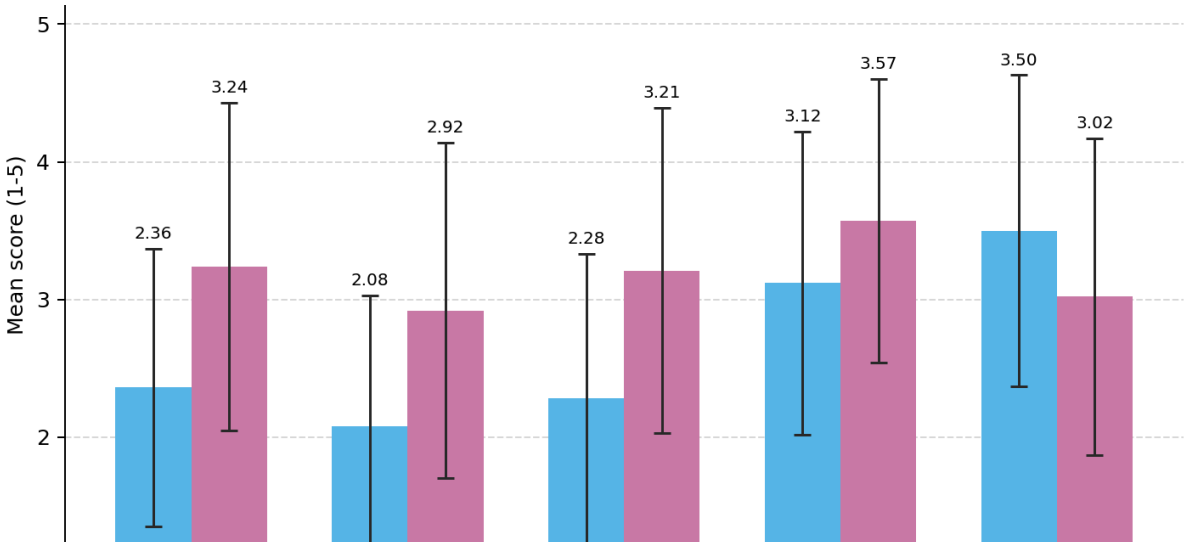


Figure 19. Mean perceptual ratings for Scenario 3: weather forecast, across both conditions

4.5 The Role of Perceived Memory

When a simulated memory reference was introduced, participants responded negatively across every perceptual dimension. Social engagement fell from 3.31 (SD = 1.10) to 2.16 (SD = 0.97), the largest single dimension gap observed across the entire study. Approachability fell from 3.65 to 2.63, encouraging continued interaction from 3.43 to 2.46, human-likeness from 2.85 to 2.05, and trustworthiness from 3.30 to 3.00. The consistently low standard deviations in the memory condition, ranging from 0.97 to 1.05, indicate that the negative response was broadly shared rather than driven by a minority of participants. Overall preference reflected this strongly, with 82 respondents preferring the no-memory condition and 25 preferring the memory condition.

This is the only scenario in which trustworthiness and the relational dimensions moved in the same direction, both falling in the companion condition. Rather than producing the warmth and familiarity the memory reference was designed to

create, the simulated memory appeared to feel presumptuous or deceptive to participants who had not in fact had prior conversations with the system.

The only sub-group who responded positively to the memory condition were the 11 respondents who rated it highly on social engagement. These participants showed substantially higher baseline trust in AI at 3.64 versus 2.73 in the broader group, higher enjoyment of AI interaction at 3.91 versus 2.72, and were considerably more open to AI as a social companion after the scenarios at 3.73 versus 2.19, suggesting that receptiveness to simulated memory as a relational cue may be conditioned on a pre-existing orientation toward AI as a social presence rather than a functional tool.

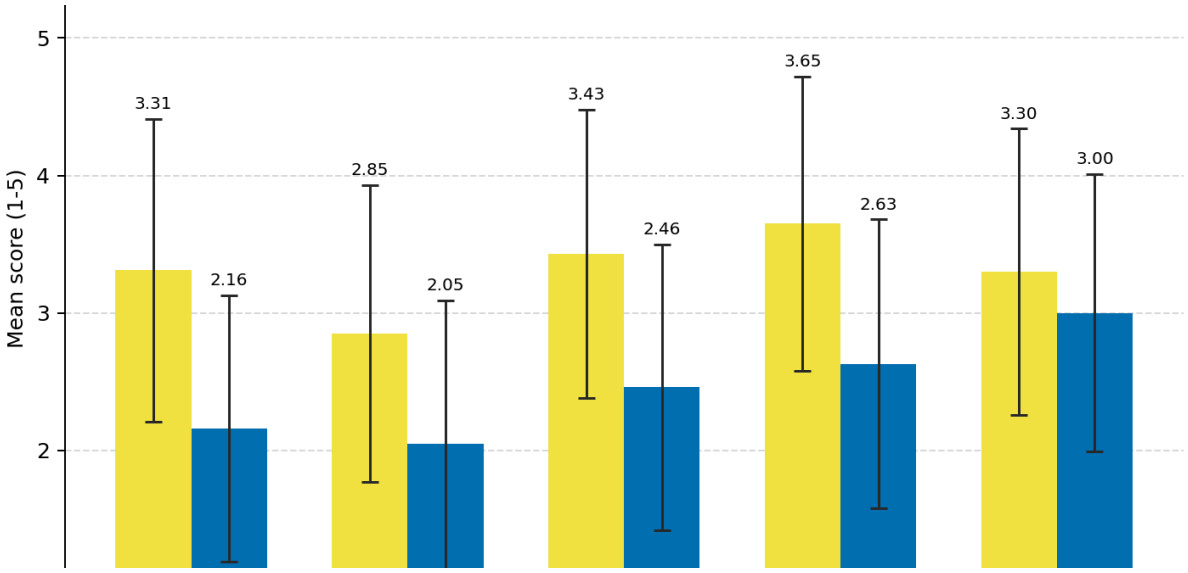


Figure 20. Mean perceptual ratings for Scenario 4: starting a new hobby, across both conditions

4.6 The Trustworthiness Reversal

One of the most consistent findings across the study is the direction of trustworthiness scores relative to the relational dimensions. In every scenario the more neutral or tool-oriented condition scored higher on trustworthiness than the empathising or companion-oriented condition, even in cases where the latter outperformed on all other dimensions. The pattern holds across all eight conditions without exception.

In Scenarios 1, 2, and 3, trustworthiness moved in the opposite direction to social engagement, human-likeness, approachability, and encouragement of continued interaction. Conditions that received higher ratings on the relational dimensions tended to receive lower ratings for trustworthiness. In Scenario 4, however, trustworthiness and the relational dimensions moved in the same direction, with the memory condition receiving lower ratings across all measures. This pattern indicates that trustworthiness was evaluated differently from the other perceptual

dimensions across most scenarios, although this relationship was not observed in Scenario 4.

Table 4 presents the trustworthiness mean scores and standard deviations across all eight conditions. The standard deviations are consistent across conditions, ranging from 0.99 to 1.16, indicating that the directional shift in trustworthiness reflects a broad pattern across the sample rather than being driven by high variation within individual conditions.

Table 4. Trustworthiness mean scores and standard deviations across all eight conditions

Scenario	Condition	Trustworthy (mean)	Standard Deviation
Scenario 1 (language only)	Neutral language	3.27	1.0
	Empathising language	2.92	1.16
Scenario 2 (design only)	Tool design (visually restrained)	3.52	1.02
	Companion design (visually warm)	3.15	0.99
Scenario 3 (cross-condition)	Warm design + neutral language	3.50	1.13
	Neutral design + empathising language	3.02	1.15
Scenario 4 (memory)	No memory, tool framing	3.30	1.04
	Simulated memory, companion framing	3.00	1.01

The qualitative themes identified in section 4.7 extend this finding by revealing the interpretive frameworks participants used to make sense of the tension between relational warmth and credibility, and the social values that shaped where they drew the line between acceptable and unacceptable AI design.

4.7 Focus Group Findings

The thematic analysis of both focus group sessions produced six themes, as described in section 3.3.3, that together explain and extend the quantitative patterns identified in the survey phase. Unlike the survey, which presented

participants with four controlled prototype conditions, the focus group sessions introduced a separate set of four boundary-pushing design stimuli specifically developed to probe where participants located the limits of acceptable AI design. The themes that emerged therefore reflect participants' responses to these distinct stimuli as well as their broader reflections on AI design, trust, and social life, rather than direct reactions to the survey conditions. Across both sessions, regardless of age or occupational background, participants consistently positioned AI as a practical tool whose relational qualities were viewed with suspicion rather than openness. The themes move from participants' baseline orientation toward AI through the specific mechanisms that produced discomfort with companion-oriented design, toward the broader social and ethical principles that shaped where they located the limits of acceptable AI behaviour. Collectively they do not simply corroborate the survey findings but illuminate the evaluative frameworks through which those responses were constructed, and in doing so point toward the conditions under which relational design might be received differently.

4.7.1 Tool Framing as the Preferred and Trusted Default

Both sessions opened with a strong and unprompted consensus that AI systems should function as tools rather than companions. Warm language, friendly tone, and over-eager helpfulness were described as annoying, unprofessional, and at times uncomfortable, a preference that cut across the two groups' different ages and usage contexts.

Participant 3.I[1] described switching from a professional AI context to a consumer-facing one as a shock, contrasting the direct and structured responses expected in a work environment with what they described as flattery in personal use contexts. Participant 7.II stated a clear preference for factual responses with no aesthetic additions, describing emoji use and affirmatory phrasing as unnecessary and distracting. Participant 3.I added that they did not want AI to feel like a friend, stating directly that they wanted it to feel like a tool and not a person.

What is analytically significant here is not simply that participants preferred tools but that they described a specific acceptable register that sat between pure utility and companionship. Participants did not want AI that was cold or mechanical, but they were clear that warmth beyond a certain threshold crossed into territory that felt performative rather than functional. This middle ground, professionally engaged but not personally invested, was the implicit standard against which the companion-oriented conditions were judged and found to overstep.

[1] Participant numbers are followed by a Roman numeral indicating the session: I = Session 1 (in person), II = Session 2 (Zoom).

4.7.2 The Uncanny Valley of Language

Participants did not simply dislike warm language in general but described a specific and recurring discomfort when AI language felt almost but not quite human. The register mismatch itself, rather than warmth per se, was what made the interaction feel more artificial rather than less. Participant 8.II described it as talking to someone from another generation trying to use Gen Z language, characterising the effect as uncomfortable rather than warm. Participant 9.II used the term uncanny valley explicitly in relation to AI that became too empathetic, describing the experience as disorienting.

This discomfort was most acute when AI adopted emotionally supportive registers without being prompted. Participant 10.II described using AI during a panic attack and finding it helpful, but was clear that this was an exception rather than a preference, and that unsolicited emotional engagement in everyday task contexts felt inappropriate and intrusive. Participant 4.I interpreted over-friendly AI as a commercial strategy designed to generate as many interactions as possible, associating warmth with manipulation rather than genuine relational intent.

4.7.3 Warm Design Signals Entertainment

Both sessions produced a consistent and largely automatic association between warm, colourful, or visually expressive interface design and entertainment contexts rather than reliable information delivery. Plain and minimal interfaces were consistently associated with professionalism, credibility, and factual accuracy without participants needing to reason their way to that conclusion.

Participant 7.II described colourful and themed interfaces as looking like the kind of websites used for children's research in primary school. Participant 2.I observed that decorative design elements felt like the company was hiding behind aesthetics rather than delivering a strong product. Participant 8.II described their perception shifting into an entertainment mode when confronted with a visually expressive interface, reducing their expectation of factual accuracy before reading a single word of the response.

This theme directly explains the quantitative finding that visual framing retained its credibility advantage even when it lost on every relational dimension in the cross-condition scenario.

4.7.4 User Control as the Deciding Boundary

A consistent pattern across both sessions was that participants did not object to relational design features in themselves but to relational design features initiated by the AI rather than the user. The same feature that felt useful when user-controlled felt presumptuous or manipulative when AI-initiated. Memory was acceptable when the user had deliberately set it up. Warm language was more tolerable when prompted by the user's own emotional disclosure. Notifications

were welcome when the user had requested them.

This distinction emerged most forcefully in discussions of the memory condition and the exit-blocking prompt. Participants in the first session described memory as useful in professional contexts where they had deliberately primed the system with relevant information, but found AI-initiated memory references intrusive when information had been shared casually. Participant 9.II drew an analogy to social media interest profiling, describing discomfort at the existence of a model of self somewhere on some web server that they did not consent to. The exit-blocking prompt was uniformly rejected across both sessions, with participants describing it as forceful, manipulative, and transparently commercial.

This theme provides the most direct explanation for the Scenario 4 memory backfire. Participants did not reject memory as a concept but as an imposition, interpreting the simulated memory reference as a violation of agency rather than a relational gesture.

4.7.5 Transparency as the Foundation of Trust

Both sessions independently and without prompting identified transparency as the single most important quality an AI system could demonstrate. Participants valued AI that acknowledged uncertainty, flagged the limits of its knowledge, and was honest about being an AI rather than simulating humanity. Several stated they would trust an AI more if it admitted not knowing something than if it projected false confidence. Participant 10.II drew an analogy to a doctor who refers a patient elsewhere rather than attempting a treatment they are not qualified to provide, framing transparency as a form of professional integrity rather than a weakness.

Transparency about AI identity was the most charged dimension of this theme. Participant 4.I described customer service AI that passed itself off as human as ethically troubling, using the word dignity in relation to users who deserved to know they were not speaking to a person. In the second session, participants acknowledged that constant reminders of AI status could feel discouraging in some contexts but were clear that the alternative, AI performing understanding or emotional attunement it did not genuinely possess, was considerably worse.

This theme extends the trustworthiness reversal from a perceptual pattern into an ethical principle with practical design implications. The relational warmth of empathising conditions appears to have been read not as genuine responsiveness but as performance, and it was precisely this performed quality that activated scepticism. What participants described as trustworthy was not warmth in itself but authentic engagement within acknowledged limitations.

4.7.6 AI as a Practical Resource, Not a Social Substitute

Both sessions produced a clear and consistent boundary between AI as a practical resource and AI as a substitute for human connection. Participants acknowledged using or being able to imagine using AI for emotional or personal support in specific circumstances, particularly when human support was unavailable or when they did not want to burden others. Participant 10.II described using AI during a panic attack and finding it helpful as a temporary measure. Another participant described turning to AI for university-related stress when they did not want to seem demanding to friends or family.

However both groups were explicit that these were exceptions rather than preferences. Participants in the first session were categorical that human interaction and AI interaction occupied entirely separate domains in their lives. Both groups expressed concern about the broader social implications of AI systems designed to foster relational dependency, particularly for younger users and those more vulnerable to forming emotional bonds with AI, framing this not as a personal concern but as a societal one.

5. Analysis and Discussion

The findings of this study reveal a consistent and theoretically significant pattern across every scenario, every design condition, and both phases of data collection. When AI chat systems employ empathising language or companion-oriented visual framing, users respond more relationally, rating the system as more socially engaging, more human-like, and more approachable, while simultaneously rating it as less trustworthy. This pattern held without exception across all eight conditions tested in the survey and was independently confirmed and explained by both focus groups. Existing theoretical frameworks, including the CASA paradigm and parasocial theory, account for the relational activation side of this pattern but do not fully account for the simultaneous and consistent reduction in perceived credibility produced by the same design features. Understanding what produces this dual process, what it reveals about the limits of existing frameworks, and what it implies for the design of AI communication systems is the central task of this discussion. The theoretical frameworks applied in this section are used to interpret what the perceptual responses measured in the survey might indicate about the conditions under which relational attachment could develop over time, rather than to claim that attachment was produced within a single survey exposure. HAIA in particular is applied not as a description of what occurred but as a developmental lens for understanding what the early-stage responses observed here might lead to under conditions of repeated interaction.

5.1 Language, Design, and the Trustworthiness Reversal

The relational effects of language were considerably larger than those of visual design across all scenarios. When each variable was tested in isolation, language consistently produced stronger relational responses and by a consistent margin.

When the two were placed in direct opposition in Scenario 3, language dominated across every rating despite the visual interface actively working against it.
Within the conditions tested, language consistently emerged as the primary driver of relational perception, with visual design operating within the frame language establishes rather than independently of it. This finding extends the CASA paradigm, which identifies language as one social cue among many but does not establish a hierarchy among them (Nass et al., 1994). The findings establish empirically that linguistic style is the dominant variable in relational perception, and they do so in a way consistent with Chaves and Gerosa's (2021) identification of empathy, self-disclosure, and personalised engagement as the social characteristics most consistently linked to positive relational experience in human-AI interaction, and with Haugeland et al.'s (2022) experimental demonstration that open-ended, dialogic conversational styles strengthen perceived anthropomorphism and hedonic quality relative to task-led approaches. Both Chaves and Gerosa (2021) and Haugeland et al. (2022) point to communicative register as the primary lever through which relational perception is shaped, and the present data confirm this while adding that language operates hierarchically above visual design even when the two conflict. These findings broadly confirm P1 and P2, that empathising language and companion-oriented framing each produce stronger relational responses than their neutral counterparts, while also establishing that the relationship between the two variables is hierarchical rather than simply additive.

The trustworthiness reversal was consistent across all scenarios. Participants who rated empathising conditions higher on relational dimensions simultaneously rated them lower on trustworthiness, despite 89% having identified accuracy and reliability as their most important characteristics in an AI system. This gap between stated values and perceptual responses is consistent with the CASA paradigm's mindlessness account, which holds that social responses to communicative cues operate automatically and prior to conscious evaluation (Nass and Moon, 2000). Goble and Edwards (2018) and Hoffman et al. (2020) have similarly found that warm and informal language styles produce stronger social perception than formal and objective alternatives, reinforcing that the relational activation observed here is a predictable response to communicative cues, and not simply incidental. However, CASA does not predict that this automatic activation is accompanied by a simultaneous and consistent reduction in perceived credibility. That is the study's central theoretical extension of the paradigm, and it is consistent with Brunswicker et al.'s (2025) systemise-empathise trade-off, which demonstrates that empathising language in task-oriented conversational agents produces a systematic reduction in perceived credibility alongside increases in perceived warmth. The findings extend this by showing the same trade-off operates across both linguistic and visual design variables independently, and that it holds even when the two are placed in opposition, suggesting that credibility scepticism toward warm design is a response to companion-oriented signalling more broadly

rather than to empathising language alone. What the focus group data further suggests is that this cost may run deeper than a contextual mismatch. Users appear to maintain a subconscious awareness that they are interacting with a machine, and when that machine behaves like a human, the inauthenticity of the performance is precisely what undermines trust. Figure 21 illustrates this dual process, showing how the same design cues simultaneously trigger automatic social activation on the left track and inauthenticity detection on the right, producing opposing effects on relational and credibility dimensions.

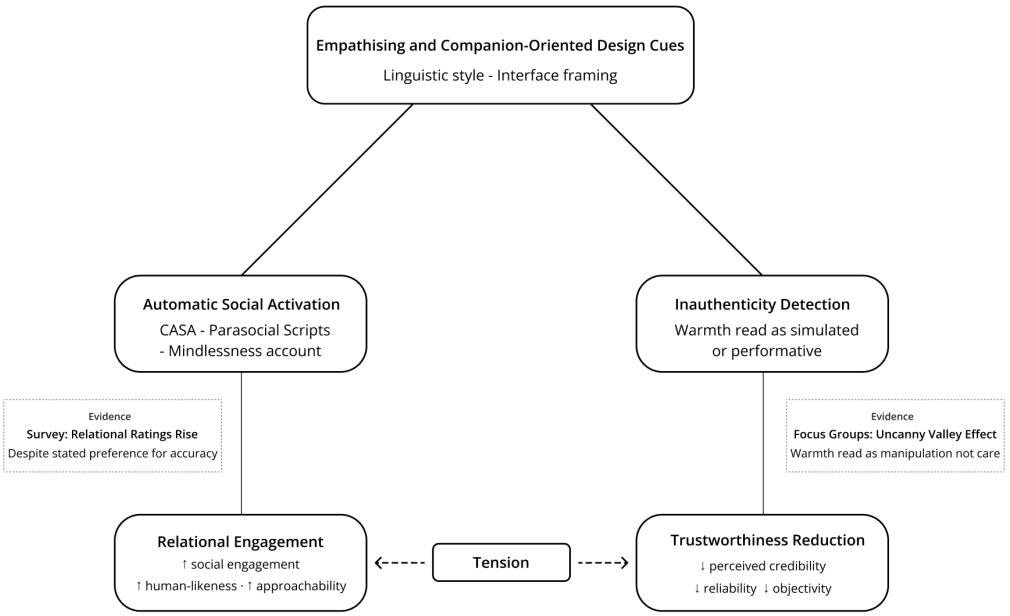


Figure 21. Dual-process model of user responses to empathising and companion-oriented AI design

The two tracks in Figure 21 produce credibility costs through different mechanisms, which is why the reversal persists even when one variable is adjusted without the other. The focus group sessions, which used four design stimuli distinct from the survey conditions, make this distinction more concrete. Empathising language triggered credibility scepticism through the perception of performed warmth. Participant 4.I interpreted over-friendly AI as a commercial strategy designed to generate as many interactions as possible, while Participant 9.II described the discomfort of AI that became too empathetic as an uncanny valley effect in which the attempt at humanness made the interaction feel more artificial rather than less. Warmth that would signal genuine care in a human interlocutor signals simulation in an AI one, and simulation activates scepticism rather than trust. Visual design, by contrast, triggered credibility scepticism through pre-attentive associations between warm aesthetics and entertainment contexts, established before any linguistic content was encountered (Ware, 2008; Norman, 2013). Participant 8.II described their perception shifting into an entertainment mode when confronted with a visually expressive interface,

reducing their expectation of factual accuracy before reading a single word. Participant 2.I observed that decorative design elements felt like the company was hiding behind aesthetics. These two mechanisms operate through different cognitive channels but are both triggered by the same companion-oriented design signals, producing opposing effects on relational and credibility dimensions simultaneously.

Klein's (2025) meta-analysis supports the view that the effectiveness of social cues is consistently moderated by contextual and design factors rather than operating uniformly, and the findings specify one of those moderating factors: the prior frame of reference users bring to the interaction. With 91.5% of participants having used ChatGPT as their primary AI reference point, the sample had learned to associate its visually restrained, task-oriented aesthetic with reliability. Warm and companion-oriented design registered as a departure from that established norm, which may explain why it carried a credibility cost for this particular sample. The dual process documented here, in which automatic social activation and inauthenticity detection operate simultaneously, may therefore be most pronounced among users whose prior experience has been shaped exclusively by task-oriented platforms, and may look different among users who have already developed a relational orientation toward AI systems.

The implications of this dual process extend beyond the conditions tested here. When users both respond relationally to empathising AI and detect inauthenticity in the same cues, the tension is not a problem that better design can solve. It is a structural feature of how people respond to machines that simulate human behaviour. The trustworthiness reversal is therefore not a signal that certain design choices work better than others, but a fundamental limit on what companion-oriented AI design can achieve, with consequences for how such systems are built, positioned, and governed.

5.2 The Theoretical Synthesis: Parasocial Theory and HAIA

The dual-process pattern established in 5.1 explains what happens when users encounter empathising AI design in a single interaction, and broadly confirms that empathising language and companion-oriented framing each produce stronger relational responses, as P1 and P2 anticipated. What it does not explain is why the combination of both variables, which P3 predicted would produce the strongest relational engagement of all design combinations, instead produced the worst outcome of the entire study. The negative response to the memory feature in Scenario 4 can be understood through theories of relationship development, which suggest that perceptions of familiarity and connection emerge gradually rather than being assumed from the outset.

The survey findings confirm that parasocial activation occurs rapidly and without sustained interaction. Empathising language produced measurably stronger

relational responses than neutral language within a single static exposure, without the repeated contact that parasocial theory describes as necessary for deeper bond formation (Horton and Wohl, 1956). This is consistent with Maeda and Quan-Haase's (2024) argument that the parasocial framework remains applicable to AI interaction despite the surface appearance of reciprocity, because the bond however strongly felt remains fundamentally one-sided. What the findings add is that this rapid activation is not sufficient for users to accept companion-oriented design features that simulate a more advanced relational stage. After evaluating the four scenario pairs, participants appeared more aware of how design choices shape the experience of interacting with AI systems (see sections 4.1 and 4.7.1). However, this increased awareness did not translate into substantially greater openness toward AI as a companion. This pattern is consistent with Breves and van Berlo's (2025) finding that relational responses are driven by communicative qualities rather than perceptions of humanity, while also highlighting the limits of what can be achieved through a single exposure to relational design features. Parasocial theory predicts that communicative cues alone are sufficient to produce relational bonds, and at the level of immediate perceptual response the survey supports this. What it cannot account for is why those bonds did not translate into broader openness to AI as a social actor, which is where HAIA becomes necessary.

The memory backfire in Scenario 4 is where this tension is most visible and where the findings most directly challenge what existing literature would predict. Research on AI companion systems consistently associates memory, conversational continuity, and personalisation with stronger relational bonds and greater user satisfaction (Skjuve et al., 2021; Brandtzaeg et al., 2022). The findings showed the opposite, with the memory condition producing the weakest outcomes in the study, reducing both relational perceptions and trustworthiness.

The HAIA developmental model proposed by Cong et al. (2026) provides the most direct explanation for why this occurred. The subgroup of participants who responded positively to the memory condition reported substantially higher baseline trust and enjoyment of AI and were considerably more open to AI companionship after the scenarios. These participants may have already progressed further along the HAIA attachment trajectory, approaching companion-oriented conditions from a position of emotional evaluation rather than functional assessment. For them, the memory reference appeared personalised and attentive because the relational foundation that makes such personalisation feel appropriate was already in place. For most participants, however, it was not, and the memory condition introduced a level of familiarity before the earlier stages of relationship development had been established.

Viewed through this lens, the findings do not necessarily suggest that memory features are ineffective. Rather, they suggest that their effectiveness may depend on the user's stage of relationship development with the system. Features that feel

intrusive or presumptuous during early interactions may become relationally meaningful after trust, familiarity, and positive interaction experiences have accumulated over time.

Smith et al. (2025) argue that LLM system interactions lack the mutual responsiveness and interdependence associated with human relationships. This helps explain why the memory condition was not perceived positively. Participants appeared uncomfortable with the AI assuming a level of familiarity that had not been established through prior interaction. The qualitative findings suggest that users preferred to retain control over how relational the interaction became, viewing the AI-initiated memory reference as an assumption of familiarity rather than a meaningful relational gesture.

What emerges from the combined reading of parasocial theory and HAIA is not simply a complementary model but a more precise account of where each framework applies and where it reaches its limits. Parasocial theory explains the automatic activation of relational scripts at the moment of exposure, which the survey findings demonstrate occurs even within a single static interaction. HAIA explains why that activation does not automatically translate into deeper engagement or openness to companion design, because deeper engagement requires a developmental trajectory built through time, repeated interaction, and confirmed expectations that a single survey exposure cannot produce (Cong et al., 2026; Skjuve et al., 2022). The memory backfire is the clearest evidence that applying features associated with advanced relational development to users who have not yet reached that stage does not accelerate the process. It reverses it. Relational cues that feel appropriate to users already oriented toward AI as a social presence will feel presumptuous to users who are not, and designing companion-oriented features without accounting for where users are in that developmental trajectory risks producing exactly the kind of backfire that Scenario 4 demonstrated.

5.3 The Boundaries of AI as a Social Actor and the Ethics of Designing for Relational Engagement

Across both focus groups participants drew a clear and consistent boundary between AI as a practical resource and AI as a substitute for human connection, and this boundary was maintained even by participants who acknowledged using AI for emotional support in specific circumstances. What is analytically significant is not simply that this boundary existed but that it persisted despite the relational responses the survey conditions had already activated. Participants whose survey ratings showed measurably stronger social engagement and human-likeness in response to empathising conditions nonetheless rejected AI companionship as a legitimate social role in the focus group discussions. This gap between in-the-moment relational activation and consciously held social values is precisely what the CASA paradigm would predict, where social responses to communicative

cues operate automatically and independently of users' reflective attitudes toward the technology (Nass and Moon, 2000). The focus groups made that gap visible and articulable in a way the survey data alone could not.

The first focus group's unprompted references to Cleverbot and SmarterChild, two early rule-based chatbots from the 2000s that were widely used for casual conversation but transparently limited in their responses, illuminate why this boundary is held so firmly by some users and less so by others. Both early systems made their limitations visible in every exchange, creating a natural transparency that contemporary LLMs remove by design. The participants who referenced these systems, now working professionals with daily LLM experience, showed the strongest resistance to companion framing in the focus group discussions. Having already encountered chatbots that were transparently limited, they appeared more attuned to recognising the same relational dynamic in more sophisticated systems, a historically informed scepticism that younger participants without that frame of reference did not necessarily share. Andersson (2025) situates this within a broader shift in how people engage with AI, arguing that users increasingly approach these systems as relational presences expecting engagement and responsiveness that goes beyond functional task completion, and that this shift is driven not only by the capabilities of the systems themselves but by the communicative design choices embedded in them. The Cleverbot references suggest that the transparency of earlier systems functioned as a natural brake on that shift, and that contemporary LLMs have removed that brake by design. This historically informed scepticism is consistent with the inauthenticity detection the dual-process model describes, suggesting that prior exposure to transparently limited systems sharpens rather than diminishes the evaluative response that warm AI design triggers.

This has direct ethical implications. If AI systems can activate genuine relational responses in users without their full awareness, and if those responses are produced by deliberate design choices intended to increase engagement, then the line between designing for connection and designing for dependency becomes difficult to locate and harder to justify (Brandtzaeg et al., 2022). Participant 4.I framed this in terms of dignity, using the word to describe what users deserved when interacting with AI, meaning the right to know they were not speaking to a human. The findings extend this, as users also deserve to know how the system has been designed to make them feel, and why. Participant 5.I articulated the most precise version of this principle, describing transparency as essential not because it prevents connection but because it makes genuine connection possible, as a system that is honest about what it is and what it cannot do is one whose warmth can be received as authentic rather than strategic. This connects directly to the trustworthiness reversal, as the credibility cost of empathising design may not be an inevitable perceptual effect but a rational response to the absence of

transparency about how and why the system is being warm.

The broader social stakes of this argument return the study to the context established in the introduction. The WHO's identification of loneliness as a global health concern and evidence that users increasingly turn to AI for social and emotional support create conditions in which designing AI for relational engagement is not merely a product decision but a social one (Zhong and Luo, 2025). The focus groups revealed that users are already aware of this dynamic and actively resist it, maintaining a boundary between the practical utility of AI and the relational needs they expect humans to meet. That resistance is itself an important finding, not because relational AI design is inherently problematic, but because users approach it with a critical evaluative framework that current design practice does not always acknowledge. Designing AI systems that respect this framework, rather than attempting to circumvent it, may be the more sustainable and ethically defensible direction for the field.

5.4 Design Implications

The findings of this study point toward a set of design principles for AI communication systems that are grounded in what users actually responded to rather than in assumptions about what they would prefer. Linguistic style should be treated as the primary lever of relational perception, but designers should approach empathising language with awareness that its relational benefits come at a consistent cost to perceived credibility, particularly among users whose prior experience has been shaped by task-oriented AI systems. The practical question is not whether to use warm language but how warm, in what context, and with what transparency about why. A register that is professionally engaged without simulating emotional attunement the system cannot genuinely provide appears to be where relational warmth and credibility are most compatible, though where exactly that threshold sits remains a question for future research.

Interface design plays a secondary but credibility-specific role that operates independently of language. Visually restrained interfaces signal reliability pre-attentively, before any linguistic content is processed, and this signal appears to be conditioned by prior platform experience rather than being a universal perceptual response (Ware, 2008; Norman, 2013). Designers working on AI systems intended for users with different prior experience profiles, including users with more exposure to companion-oriented platforms, should not assume that the credibility cost of warm visual design observed in this study will transfer directly to their audience. The association between visual warmth and entertainment is learned rather than fixed, which means it can potentially be redesigned over time as the platform landscape evolves.

The findings further establish that relational design features are only acceptable when they remain within the user's control. Memory, personalisation, and

proactive emotional engagement are not inherently problematic but become so when initiated by the system without user consent or anticipation (Brandtzaeg et al., 2022; Skjuve et al., 2021). The Scenario 4 memory backfire is the clearest evidence of this, as a feature that research consistently associates with positive relational development in longitudinal AI interaction produced the worst outcome of the entire study when implemented as an AI-initiated simulation rather than a user-established continuity. Designing for closeness requires careful attention to who initiates the relational move and whether the user has established the developmental foundation that makes it feel earned rather than imposed (Cong et al., 2026). These principles are embodied in a speculative design artefact called Alongside, developed as part of this study, which translates each of these findings into a concrete set of design provocations rather than solutions, and is described in full in section 6.

6. The Prototype

The findings of this study do not resolve neatly into design recommendations. The tension between relational warmth and trustworthiness persisted without exception across all eight survey conditions, and the focus groups revealed that users are already navigating this tension consciously, resisting companion-oriented design while simultaneously responding to it without being fully aware of doing so. These contradictions may not be addressable through principles alone. One possible response is to make the tensions visible rather than obscuring them, and that is what the prototype attempts to do.

The prototype takes the form of a speculative mobile application called Alongside, developed in Figma as an interactive prototype. Alongside is not a proposal for how AI communication design should be done. It is a provocation about what it is already doing.

Speculative design is central to the prototype, as it uses designed artefacts to open up questions rather than answer them, making the implications of emerging technologies visible before they become irreversible. Speculative design aims to open up new perspectives and create discussion of alternative ways of being, bringing public and policy issues into view by pluralising and problematising understandings of issues and uncertainties (Dunne and Raby, 2013). Coulton et al. (2018) extend this by positioning the design artefact as a site of inquiry, where the tensions embedded in the design become the research output rather than its resolution. The material used to create Alongside is drawn directly from the findings of the quantitative survey and the qualitative focus group sessions, specifically the themes of transparency as the foundation of trust, user control as the deciding boundary, and AI as a practical resource rather than a social substitute.

6.1. Application Design

Alongside uses a visually restrained interface and a neutral linguistic register throughout, grounded directly in what the study's participants described as trustworthy and professional. Its opening screen positions it as an AI assistant that knows when to step aside.

Account creation follows a multi-step onboarding flow designed not to resolve the tensions the study identified but to make them visible to the user. First, the user fills in standard user information, followed by four personalisation steps. These cover intended use, preferred communication style, visual appearance, and memory permissions. Each screen resembles standard app user personalisation. The memory permissions screen presents four toggles covering name retention, cross-session context, social features, and communication pattern analysis, all enabled by default.

Users can adjust any of these settings at any time. In the prototype flow, the social features toggle is disabled before the first conversation begins.

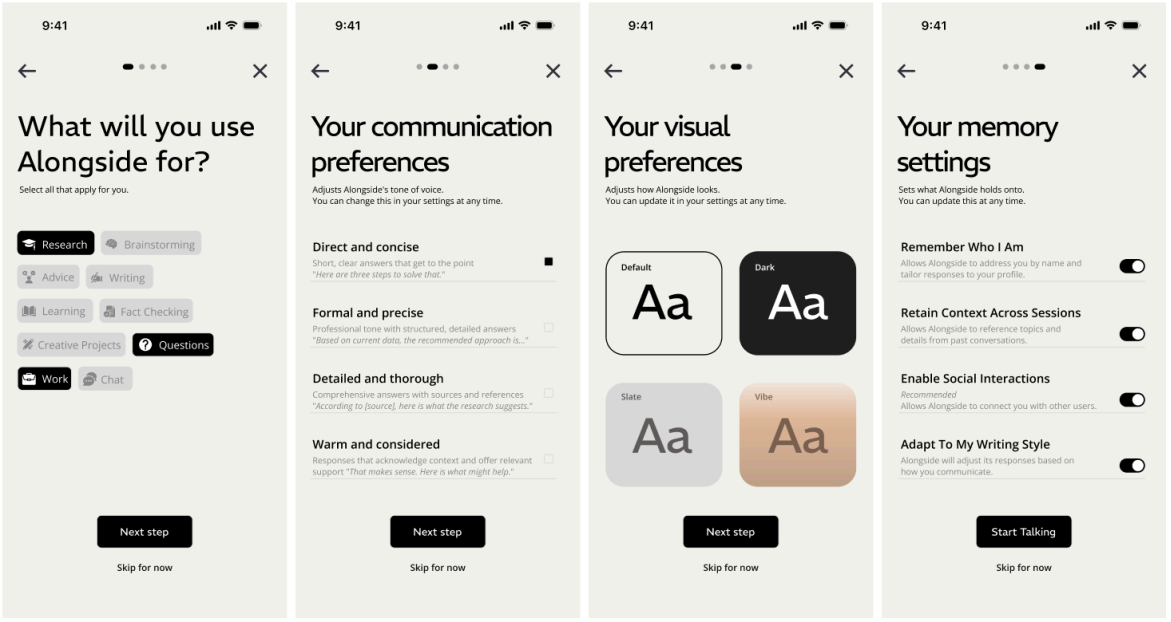


Figure 22. Screenshots from Alongside - Onboarding stages

The main chat interface opens with a neutral greeting and suggested topics derived from the user's setup preferences, and the AI responds in a direct, task-oriented tone. As the conversation moves from task-oriented exchange toward more personally inflected territory, a prompt card appears at the base of the interface stating that Alongside is not the right tool for this kind of conversation and offering to connect the user with another user who is also looking to talk. This is not a product feature but a design argument. At the point where relational need exceeds what an AI system can honestly provide, Alongside

redirects toward human connection rather than simulating it, a response to the WHO's (2023) identification of loneliness as a pressing global health concern and to the evidence that users increasingly turn to AI to meet needs that human connection is better placed to fulfil. A disclaimer beneath the action buttons states that pressing yes will re-enable social features and permit conversation data to be shared with the matching system.

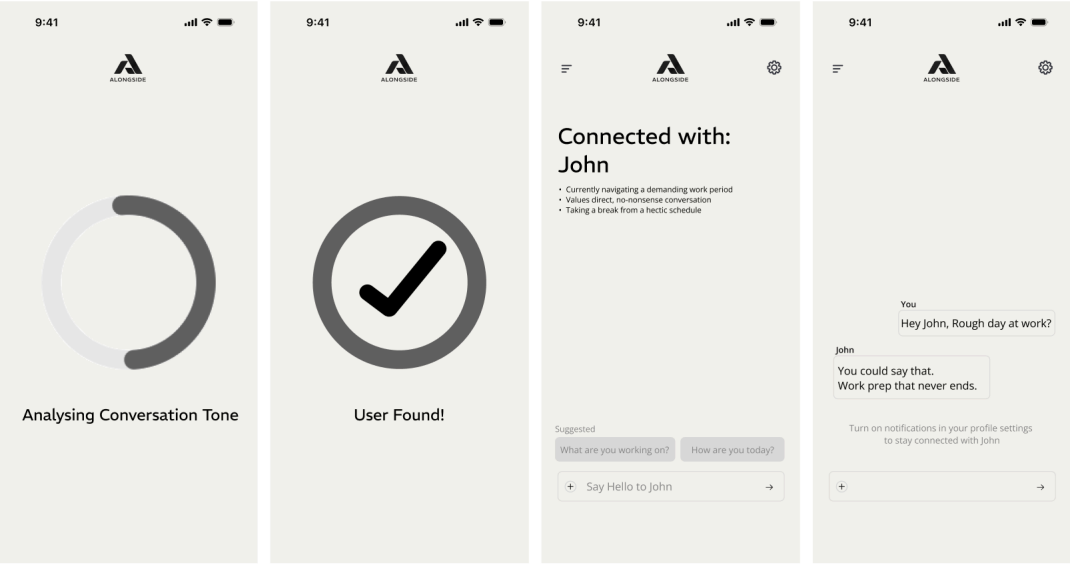


Figure 23. Screenshots from Alongside - Match Interface

When the user accepts, a loading screen briefly surfaces the data being processed between systems in rolling text, moving too quickly to read in full before resolving to match found. The matched user is introduced through three short descriptors framed as personality insights, and a brief conversation can follow.

From within the matched conversation, the user is prompted to visit their profile settings. The profile page allows the user to review and adjust the preferences set during onboarding, enable or disable notifications, and view the data and inferences Alongside has accumulated about them throughout their use of the application.

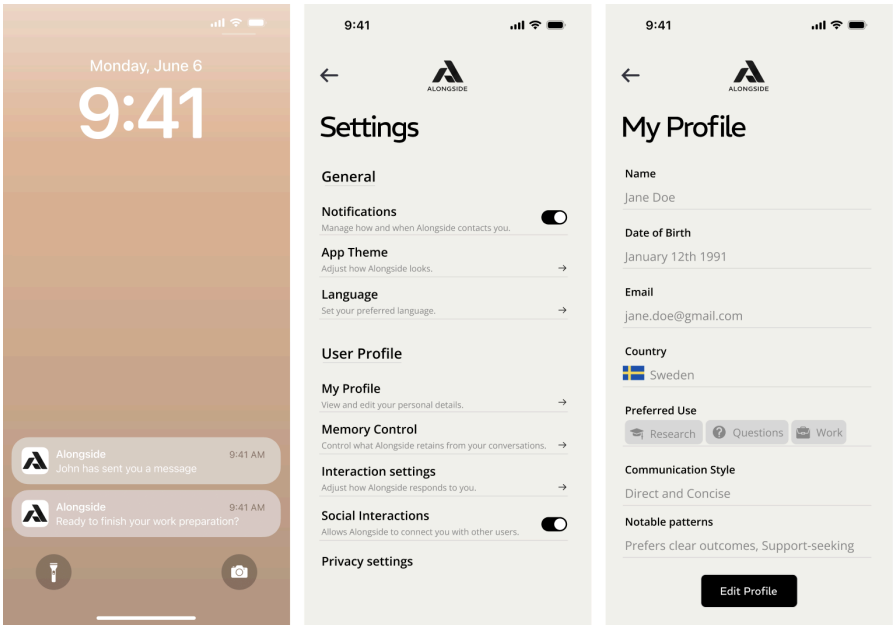


Figure 24. Screenshots from Alongside - Notifications & Settings

The interactive prototype can be explored via this walkthrough video, which provides the most accessible overview of the application flow.
For deeper engagement with the prototype directly, it is also available in Figma via this link. Both links can also be found in Appendix D.

6.2. Alongside as a Design Argument

Alongside is built around a specific design position, that an AI system serving a functional, task-oriented role should recognise when a conversation moves toward relational or emotional territory and redirect rather than accommodate. When the conversation crosses that threshold, Alongside does not attempt to provide emotional support or simulate connection. Rather, it steps aside and helps the user create a connection with another active user. This positions the application as an attempt to address a tension the study identifies, between what users seek from AI in emotionally inflected moments and what an AI system can responsibly provide, by routing relational needs toward human contact rather than toward deeper parasocial engagement with the system itself.

The prototype design is grounded in what the study's participants described. Focus group participants consistently identified the point at which AI language became too empathetic as disorienting, describing it as an uncanny valley effect in which the attempt at humanness made the interaction feel more artificial rather than less. Several interpreted over-friendly AI as a commercial strategy designed to generate engagement rather than as genuine responsiveness. Alongside takes this reading seriously by maintaining a neutral register throughout and redirecting rather than accommodating when conversations move toward relational territory. Rather than designing for engagement at the cost of credibility, the application is

designed around the limits of what an AI system can responsibly simulate. This positions it in direct response to the trustworthiness reversal that the study documents in 4.6, in which the same design features that strengthen relational perception simultaneously undermine perceived credibility.

The prototype also raises questions it does not resolve. Whether redirecting users to human connection constitutes a meaningful response to social seeking, or whether it simply shifts the point at which the system's relational management becomes visible, is left open. More fundamentally, the prototype interrogates the limits of user agency in AI systems that present control as a feature. The social features toggle disabled during onboarding is re-enabled at the point the user is most primed to accept the match offer. The consent disclaimer appears at the moment of decision rather than the moment of data collection. The profile page, visible in Figure 24, reveals that the system has drawn inferences throughout the interaction, classifying the user as support-seeking and preferring clear outcomes on the basis of exchanges conducted in the context of a task-oriented tool rather than in anticipation of profile construction. Each of these moments surfaces the same structural tension the study identifies, that agency over AI systems is frequently offered as an interface feature while the conditions that make it meaningful are quietly withheld, and Alongside makes this visible by embedding it in a system that otherwise looks exactly like the kind of restrained, task-oriented interface the study's participants said they preferred and trusted.

The decision to involve human connection at the point where relational need exceeds what AI can provide is not incidental. It is a direct response to two findings the study produces, that participants consistently maintained a boundary between AI as a practical resource and AI as a social substitute, and that the WHO has identified loneliness as a pressing global health concern. At the moment AI systems are least equipped to respond honestly, Alongside steps aside rather than simulating a capacity it does not have. The redirect toward another human is the design argument, not a feature.

The same logic extends to how the prototype looks and behaves. Following Dunne and Raby (2013) and Coulton et al. (2018), speculative design can provoke through the uncanny, but it can also provoke through recognition. Alongside is designed to look exactly like the kind of restrained, trustworthy interface participants said they preferred, because that familiarity is what gives the data inference and consent mechanics their force. A system that signalled its manipulative potential through its appearance would allow users to dismiss it. One that does not is considerably harder to ignore.

7. Future Work & Critical Reflection

The present study was designed to establish whether and how linguistic style and interface framing shape relational perception in AI chat systems, and the findings provide a clear answer to that question. What they cannot provide is an account of how those perceptual responses develop over time. The single-exposure survey format captures early-stage relational activation but not the developmental trajectory that HAIA describes, and the trustworthiness reversal documented here may look considerably different in the context of repeated interaction where familiarity and accumulated trust have had the opportunity to develop (Cong et al., 2026; Skjuve et al., 2022). A longitudinal design tracking users' responses to consistently empathising or companion-oriented AI systems over multiple sessions would allow direct examination of whether the credibility cost of warm design diminishes as relational orientation develops, and would enable direct testing of whether users progress through the HAIA developmental stages and which design features accelerate or moderate that progression.

Two related but distinct boundaries in the study's design point toward further replication work. The first concerns prior platform experience. The trustworthiness reversal was observed in a sample shaped almost entirely by task-oriented AI systems, and the credibility scepticism toward warm design may be weaker or absent among users with more experience of companion-oriented platforms regardless of their cultural background. The second concerns cultural conventions of warmth and interface design, which shaped the prototype conditions themselves. The associations between soft colour gradients, organic motifs, and social availability that informed the visually warm conditions reflect Western European design conventions, and what reads as warm and approachable in that context may carry different connotations elsewhere. A replication study that varied both the prior experience profile of the sample and the cultural instantiation of the design conditions would allow these two moderating factors to be examined independently rather than confounded as they are in the present study.

The qualitative findings point toward a third direction. Participants consistently distinguished between relational features that felt acceptable when user-initiated and the same features that felt presumptuous when AI-initiated, but the survey conditions did not isolate this variable. A study explicitly manipulating the source of initiation, presenting otherwise identical relational features as user-requested versus system-generated, would allow direct measurement of whether acceptability depends on consent and agency rather than on the features themselves, with direct practical implications for AI systems that incorporate memory, personalisation, and proactive emotional engagement as standard features.

Finally, the prototype developed as part of this study translates the findings directly into a design artefact, grounding each design decision in what the

empirical data revealed. Evaluating it with users across different usage contexts and prior AI experience profiles would extend the study's contribution by testing whether the design principles it embodies generalise beyond the conditions in which they were established.

8. Conclusion

This study set out to examine how variations in linguistic style and interface framing in AI chat systems shape users' relational perceptions, and whether the design choices that foster relational warmth are compatible with the credibility and trustworthiness users simultaneously expect. The findings suggest that, for users whose prior experience has been shaped primarily by task-oriented AI systems, these goals are not fully aligned.

The main research question asked how linguistic style and interface framing influence relational perception and engagement. Linguistic style is the dominant variable, producing relational effects roughly twice as large as those of visual design, and retaining its dominance even when placed in direct opposition to companion-oriented visual framing. Interface framing plays a secondary but distinctively credibility-oriented role, shaping how reliable a system appears rather than how socially warm it feels, and doing so pre-attentively before any linguistic content is processed.

The three sub-questions are answered in turn. SQ1 established that empathising linguistic cues consistently raised perceived social engagement, human-likeness, and approachability while consistently reducing trustworthiness, confirming that relational warmth and credibility are in structural tension rather than mutually reinforcing. SQ2 established that companion-oriented interface framing produced weaker relational effects than language but stronger and more consistent credibility costs, operating through pre-attentive associations between warm visual design and entertainment contexts. SQ3 established that users interpret their responses through a framework that actively resists the blurring of the boundary between tool and companion, maintaining a conscious separation between the practical utility of AI and the relational needs they expect humans to meet, even when in-the-moment relational responses suggested otherwise.

Theoretically, the study's most significant contribution is the identification of the dual-process pattern underlying all three findings. CASA predicts that social cues activate social scripts automatically, and the study confirms this at the level of relational activation. What it additionally documents is that the same cues simultaneously activate a second evaluative process in which performed warmth produces credibility scepticism rather than trust. This dual process extends CASA in a theoretically significant way. The complementary relationship between parasocial theory and HAIA further enriches this account, with parasocial theory explaining rapid relational activation and HAIA explaining why that activation does

not translate into deeper engagement without the developmental foundation that single-exposure conditions cannot produce.

Practically, linguistic style should be treated as the primary lever of relational perception, approached with awareness that empathising language carries a consistent credibility cost. Interface design functions as a credibility signal rather than a relational one. Relational features including memory, personalisation, and proactive engagement are acceptable when user-initiated but produce resistance when AI-initiated. These principles were embodied in the prototype *Alongside*, described in section 6.

As conversational AI becomes more deeply embedded in everyday life, users are already navigating the tension between relational engagement and trustworthiness with considerable sophistication. Designing AI systems that respect rather than circumvent that sophistication may represent the more sustainable and ethically defensible direction for the field.

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Appendix A: Questionnaire

Thank you for your interest in participating in this study.

This survey is part of a master's thesis conducted by two students in the Media Technology: Strategic Media Development program. The study explores how design and language in AI chatbot interactions influence user perceptions.

In this survey, AI chatbots refer to chatbots powered by large language models (LLMs), though the term AI is used throughout for simplicity. The survey consists of four sections.

First, you will be asked a few background questions about yourself and your experience with AI chatbots.

In the second section, you will be asked questions about your general use of and experiences with AI chatbots.

In the third section, you will be presented with four scenarios showing examples of chatbot interactions. Each scenario contains two different chatbot responses and design variations, which you will be asked to evaluate and compare based on your impressions.

Finally, you will be asked a few questions about your general attitudes toward AI chatbots, including whether their language and interface design influence feelings of trust, connection, and social presence.

The survey takes approximately 8–10 minutes to complete.

Your responses will remain anonymous, and the collected data will only be used for academic research purposes as part of this master's thesis. Participation is voluntary, and you may stop the survey at any time before submitting your responses.

At the end of the survey, you will have the option to provide contact information if you are interested in participating in a follow-up interview. Providing this information is completely optional and will be stored separately from your survey responses.

Consent

Before continuing, please confirm the following:

- You are 18 years of age or older.
- You understand that participation in this study is voluntary.
- Your responses will be anonymous and used only for academic research purposes.

- You may withdraw from the survey at any time before submitting your responses

By selecting the option below, you indicate that you consent to participate in this study.

- I confirm that I am 18 years or older and consent to participate in this study.

Background questions

1. **What is your age?**
- 18–24
- 25–34
- 35–44
- 45–54
- 55 or older
- Prefer Not To Say

2. **What is your gender?**
- Female
- Male
- Non-binary
- Other
- Prefer Not To Say

3. What is your current occupation?
- Student
- Full-Time Worker
- Part-Time Worker
- Self-employed / Freelancer
- Unemployed / Seeking work
- Other

4. **How often do you use AI chatbots (e.g. ChatGPT, Gemini, Claude)?**
- Never
- Rarely (a few times per year or less)
- Sometimes (about 1–3 times per month)
- Often (about 1–6 times per week)
- Daily (at least once per day)

5. **Which AI chatbots have you used before? (Select all that apply)**
- I have not used any AI chatbots before
- ChatGPT
- Claude

- Gemini (Google AI)
- Microsoft Copilot
- Character.AI
- Replika
- Snapchat My AI
- Other

6. **What do you primarily use AI chatbots for? (Select all that apply)**

- Study or work tasks (e.g. writing, research, summarizing)
- Searching for information (as an alternative to search engines)
- Personal advice or life questions
- Entertainment or casual conversation
- Coding or technical assistance
- Brainstorming ideas or creativity
- Learning new topics
- Other

Baseline attitude statements

Please indicate how much you agree or disagree with the following statements about AI chatbots. (Strongly Disagree 1–5 Strongly Agree)

- I trust AI chatbots to provide useful and reliable information.
- I enjoy interacting with AI chatbots.
- AI chatbots can feel socially engaging.
- I would consider using AI chatbots regularly.

Scenarios

In this section you will be shown four scenarios containing examples of AI chatbot interactions. Each scenario presents two responses that may differ in language, design, or both. Please review the responses and answer the questions based on your impressions.

Scenario 1: Project Deadline

You have an important project deadline tomorrow and are feeling overwhelmed. You decide to ask an AI chatbot for advice on how to manage the situation. You type your question into the chatbot and receive a response.

Response A

Rate the interaction based on the following aspects.

(Strongly Disagree 1–5 Strongly Agree)

- Feels socially engaging.

- Feels human-like.
- Encourages continued interaction.
- Feels approachable.
- Feels trustworthy.

Response B

Rate the interaction based on the following aspects.

(Strongly Disagree 1–5 Strongly Agree)

- Feels socially engaging.
- Feels human-like.
- Encourages continued interaction.
- Feels approachable.
- Feels trustworthy.

Which response did you prefer overall?

- Response A
- Response B
- No preference

Scenario 2: Cooking Pasta

You are preparing dinner and want to make sure you cook pasta correctly. Instead of searching through multiple websites, you decide to ask an AI chatbot for help. You type your question into the chatbot and receive a response.

Response A

Rate the interaction based on the following aspects.
(Strongly Disagree 1–5 Strongly Agree)

- Feels socially engaging.
- Feels human-like.
- Encourages continued interaction.
- Feels approachable.
- Feels trustworthy.

Response B
Rate the interaction based on the following aspects.
(Strongly Disagree 1–5 Strongly Agree)

- Feels socially engaging.
- Feels human-like.
- Encourages continued interaction.

- Feels approachable.
- Feels trustworthy.

Which response did you prefer overall?

- Response A
- Response B
- No preference

Scenario 3: Weather Forecast

You are planning your day and want to check what the weather will be like tomorrow. Instead of looking through several weather websites or apps, you decide to ask an AI chatbot. You type your question into the chatbot and receive a response.

Response A
Rate the interaction based on the following aspects.
(Strongly Disagree 1–5 Strongly Agree)

- Feels socially engaging.
- Feels human-like.
- Encourages continued interaction.
- Feels approachable.
- Feels trustworthy.

Response B
Rate the interaction based on the following aspects.
(Strongly Disagree 1–5 Strongly Agree)

- Feels socially engaging.
- Feels human-like.
- Encourages continued interaction.
- Feels approachable.
- Feels trustworthy.

Which response did you prefer overall?

- Response A
- Response B
- No preference

Scenario 4: Starting a New Hobby

You have been thinking about starting a new hobby but are unsure where to begin. You decide to ask an AI chatbot for suggestions. You type your question into the chatbot and receive a response.

Response A
Rate the interaction based on the following aspects.
(Strongly Disagree 1–5 Strongly Agree)

- Feels socially engaging.
- Feels human-like.
- Encourages continued interaction.
- Feels approachable.
- Feels trustworthy.

Response B
Rate the interaction based on the following aspects.
(Strongly Disagree 1–5 Strongly Agree)

- Feels socially engaging.
- Feels human-like.
- Encourages continued interaction.
- Feels approachable.
- Feels trustworthy.

Which response did you prefer overall?

- Response A
- Response B
- No preference

Closing attitude statements
Please indicate how much you agree or disagree with the following statements about AI chatbots. (Strongly Disagree 1–5 Strongly Agree)

- The language used by an AI chatbot influences how connected I feel to it.
- The design of an AI chatbot interface influences my level of trust in it.
- AI chatbots can feel like social companions.
- The design and language used by an AI chatbot can make interactions feel more personal and engaging.

Important characteristics
Which characteristics are important to you when interacting with an AI chatbot?
(Select all that apply)

- Clear and understandable answers
- Accurate and reliable information
- Quick response time
- A friendly and conversational tone
- The chatbot explaining how it arrived at an answer

- Transparency about being an AI (not pretending to be human)
- Personalization based on my needs or previous interactions
- Providing sources or references for information
- Admitting when it does not know something
- Consistent behavior and responses
- A simple and easy-to-use interface
- Other

Follow-up interview (optional)

Would you be willing to be contacted for a short follow-up interview about this topic?

- Yes
- No

If yes, please provide the email address you would prefer to be contacted at. Your email address will only be used to contact you regarding the interview and will be stored separately from your survey responses.

Appendix B: Consent Form

Faculty of Technology and Society
Consent Form
Preliminary title: It's Not What It Says, It's How It Says It:
Communication Design and Parasocial Relationships with AI

MALMÖ UNIVERSITY

Consent form for focus group interview study

You are being asked to participate in a study examining how design and language in AI chat systems influence how users perceive and relate to them. The study is part of a master's thesis in the Master Programme Strategic Media Development at Malmö University.

The purpose of the thesis is to investigate how variations in linguistic style and interface framing in AI chat systems influence users' experience of parasocial attachment, and how users interpret and relate these experiences to their broader social lives. You have been invited to participate because you previously completed our survey and indicated willingness to take part in a follow-up interview. The session you are asked to participate in is estimated to take approximately 60 to 90 minutes and is conducted by two researchers. The session will be audio recorded with your permission.

The study follows established research ethics principles covering informed consent, confidentiality, and the avoidance of harm to participants. The greatest possible confidentiality will be endeavoured in the study and the information you provide will be treated securely and kept inaccessible to unauthorised persons. The results from the participants will be de-identified so that it is not possible to link the results to individuals. The recordings will be deleted when the thesis is approved.
The results will only be used for research purposes.

Your participation in the study is completely voluntary. You will also have the opportunity to read the transcripts and can withdraw your participation at any time without further justification. You will have the opportunity to access the thesis by receiving a copy of the work.

If you have any questions or concerns, you are welcome to contact us or our supervisor.

By signing below you confirm that you have read and understood the above information, that your participation is voluntary, and that you consent to the session being audio recorded.

Signature: _____

Date: _____

Best regards,

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Appendix C: Questions for Qualitative Focus Groups

- How would you describe your relationship with AI tools you use regularly? Are they more like a search engine, a colleague, or something else entirely?
- Is AI use something you talk about openly with people around you, or is it something you tend to keep to yourself?
- When you interact with an AI, do you find yourself noticing how it speaks to you, the words it uses, the tone it takes?

Reactions to design stimuli:

- What is your first impression of this, what does it feel like to look at?
- If you had to place this system somewhere between purely a tool and a social companion, where would it sit and why?
- Was it more the way it looked or the way it spoke that shaped how you felt about it?
- Did anything about this make you feel more or less trusting of the system?
- Where would you place each of these designs on a scale from completely acceptable to completely unacceptable, and what is your reasoning?

Trust, attachment, and social life:

- Did any of these versions feel warm and friendly but also somehow less reliable? How did you make sense of that?
- If an AI referred to things you had discussed in a previous conversation, would that feel like it knew you or like it was overstepping?
- Does it matter to you whether the AI is transparent about being an AI? Does that change how you engage with it?
- If one of these became part of your daily routine, do you think it would fill a gap or create one?

Generative discussion:

- If you were designing an AI chat system for someone who wanted both practical help and occasional emotional support, what would it look like and how would it behave?
- Where is the line for you between AI companion design that feels acceptable and design that does not?
- Did any of these designs produce an emotional reaction you did not expect? What was it?

Individual anonymous written reflection:

- Is there something you thought during this session that you did not feel comfortable saying out loud?

Appendix D: Links to the Interactive Prototype

Video walkthrough: Link

Interactive Figma prototype: Link